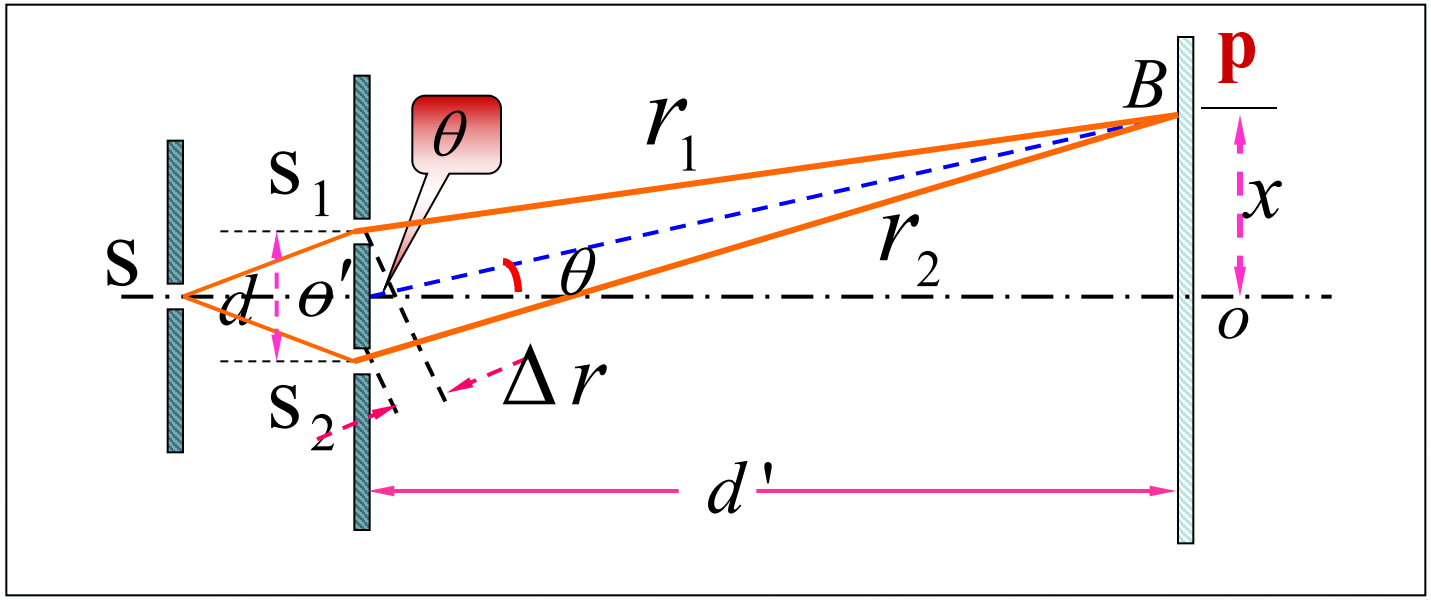
相干条件： 频率相同
振动方向相同
相遇点有恒定的相位差

相干光(coherent light)：能产生干涉现象的光。

相干光源(coherent source)：能产生相干光的光源。



$$\Delta r \approx d\sin\theta = \begin{cases} \pm k\lambda & \text{加强} \\ \pm(2k+1)\dfrac{\lambda}{2} & \text{减弱} \end{cases} \qquad k = 0,1,2,\cdots$$

$$d\frac{x}{d'} = \begin{cases} \pm k\lambda & \text{明纹} \\ \pm(2k+1)\dfrac{\lambda}{2} & \text{暗纹} \end{cases} \qquad k = 0,1,2,\cdots$$

干涉条纹在屏幕上的分布

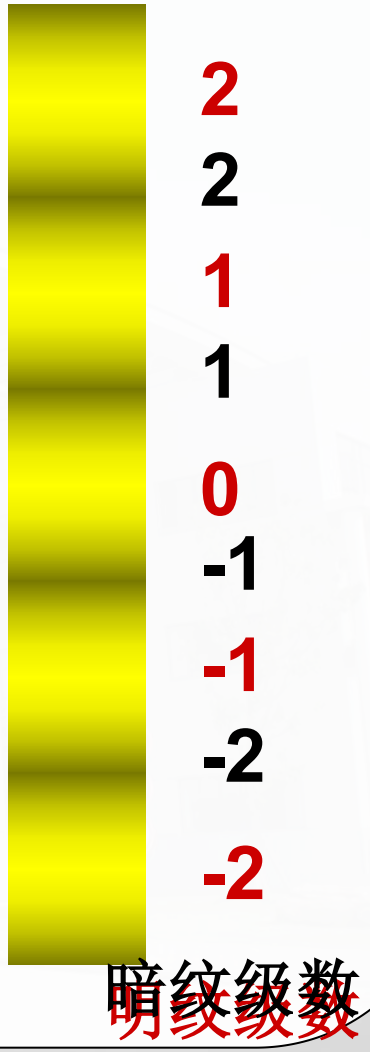
明纹： $x = \pm k\dfrac{d'\lambda}{d} \quad (k = 0,1,2,\cdots)$

暗纹： $x = \pm(2k-1)\dfrac{d'\lambda}{2d} \quad (k = 1,2,\cdots)$

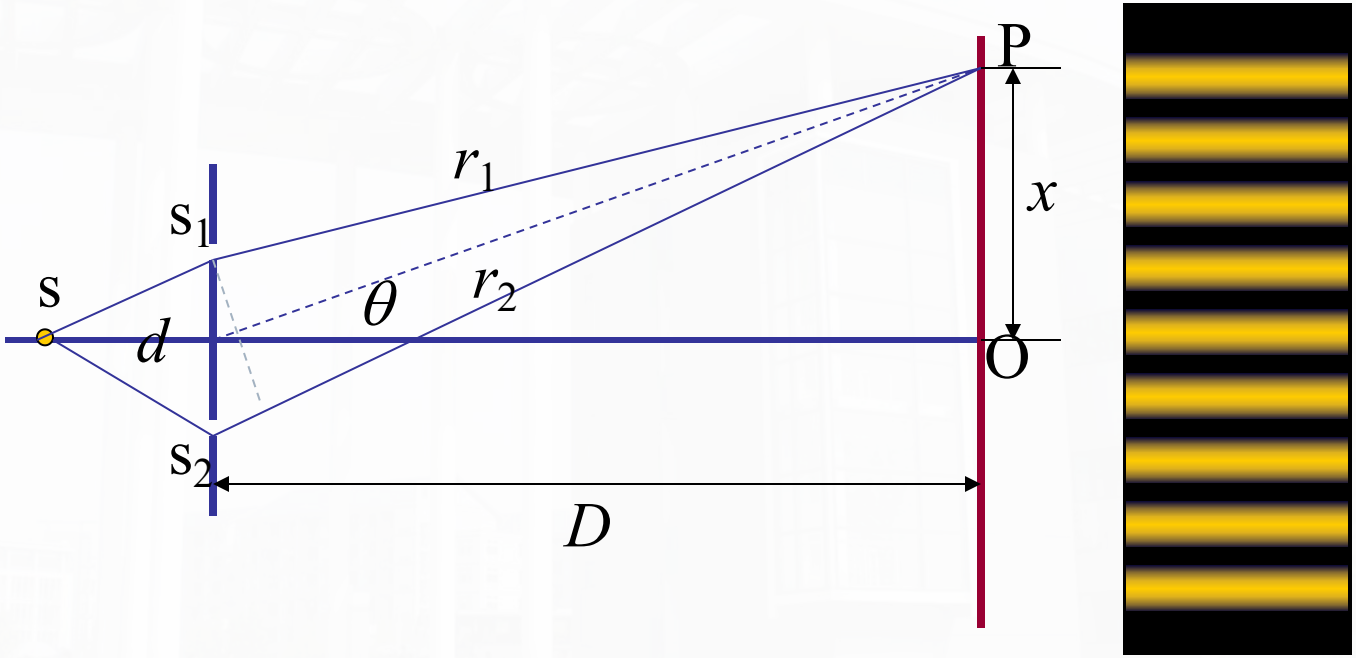
屏幕中央（k = 0）为中央明纹

相邻明纹或暗纹的间距

$$\Delta x = x_{k+1} - x_k = \frac{d'}{d}\lambda$$

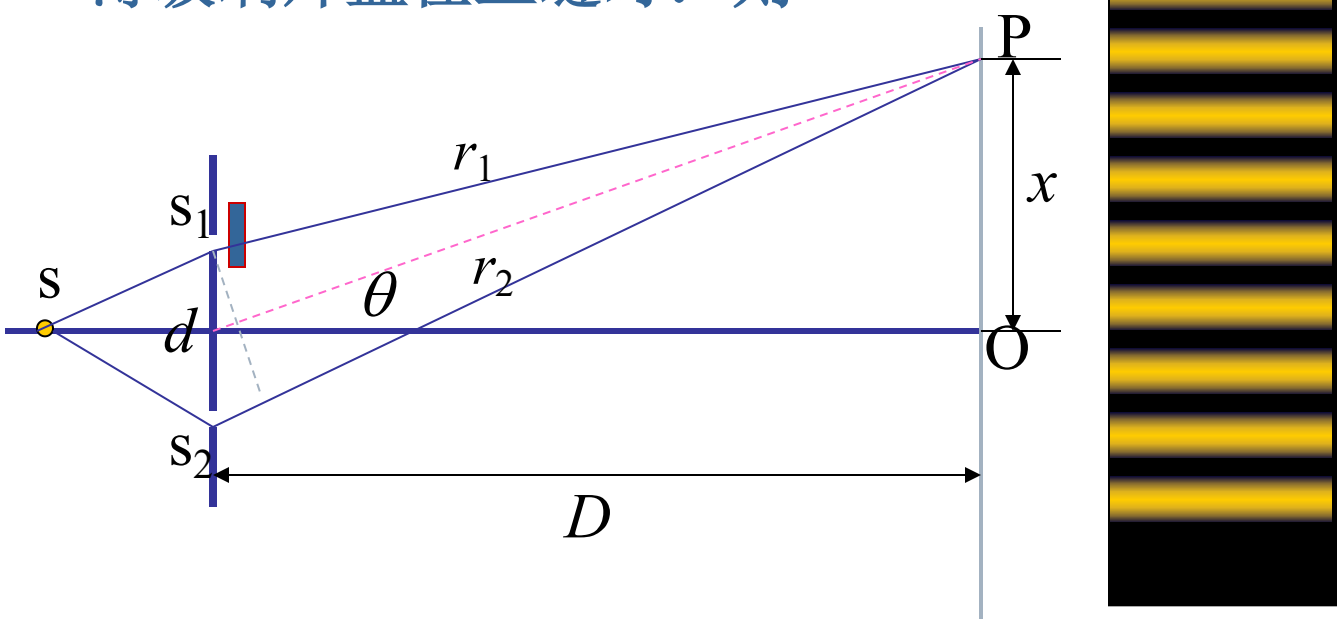


- 在缝后加一薄玻璃片，观察条纹的移动



• 在缝后加一薄玻璃片，观察条纹的移动

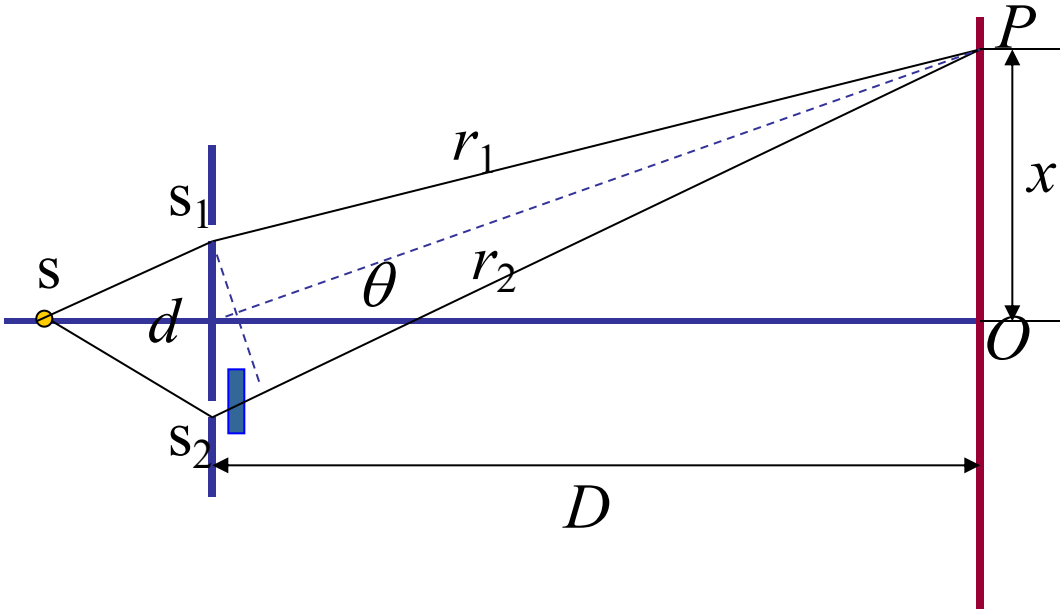
薄玻璃片盖住上缝时：则

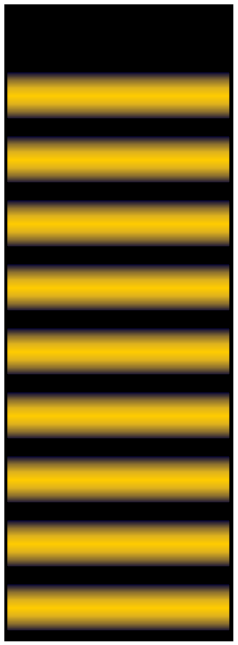


中央明纹上移

- 在缝后加一薄玻璃片，观察条纹的移动

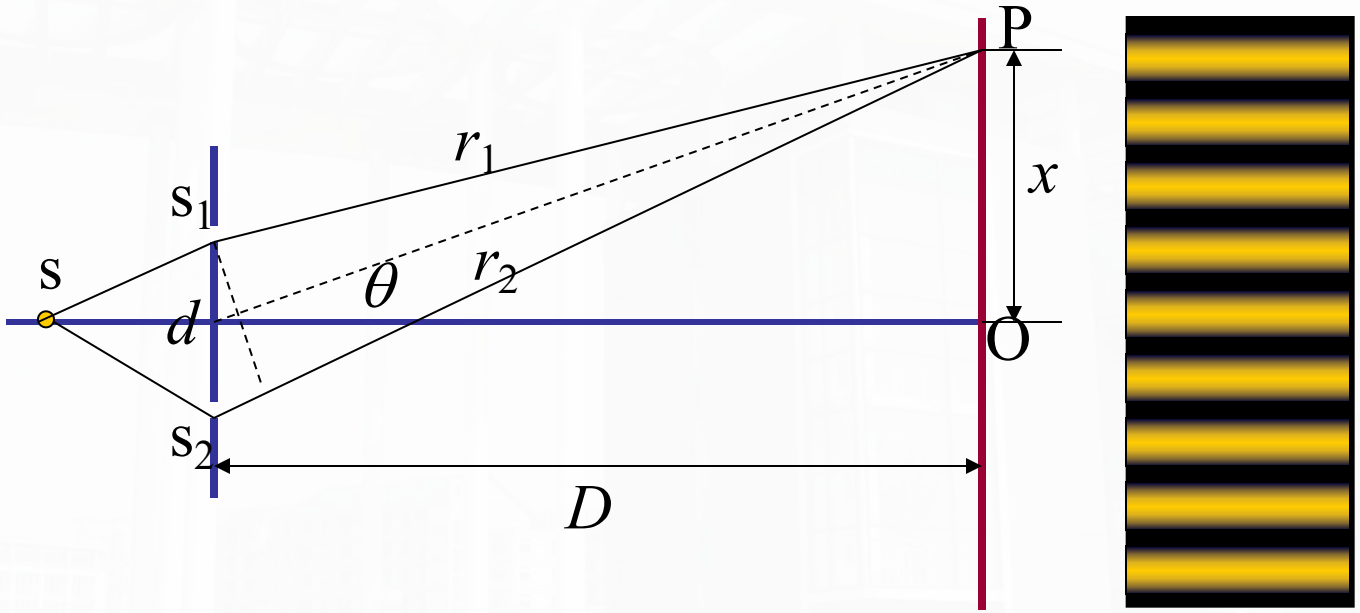
薄玻璃片盖住下缝时：则





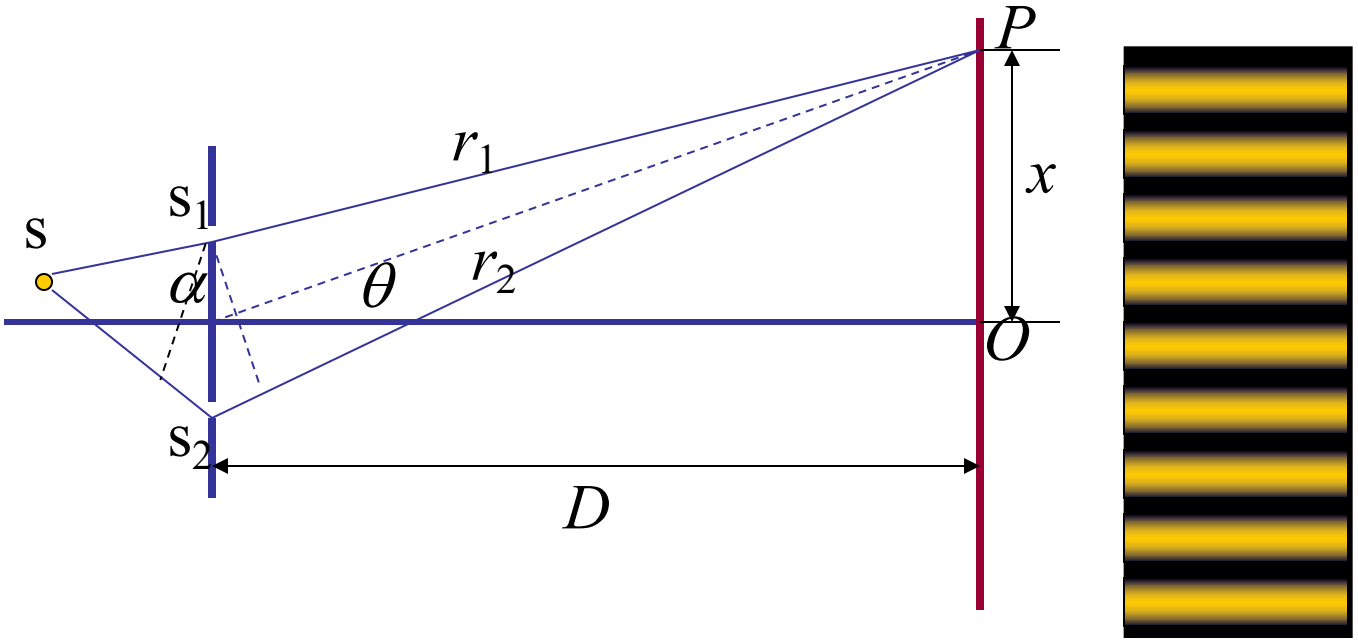
中央明纹下移

- 上下移动光源时，观察条纹的移动



- 上下移动光源时，观察条纹的移动

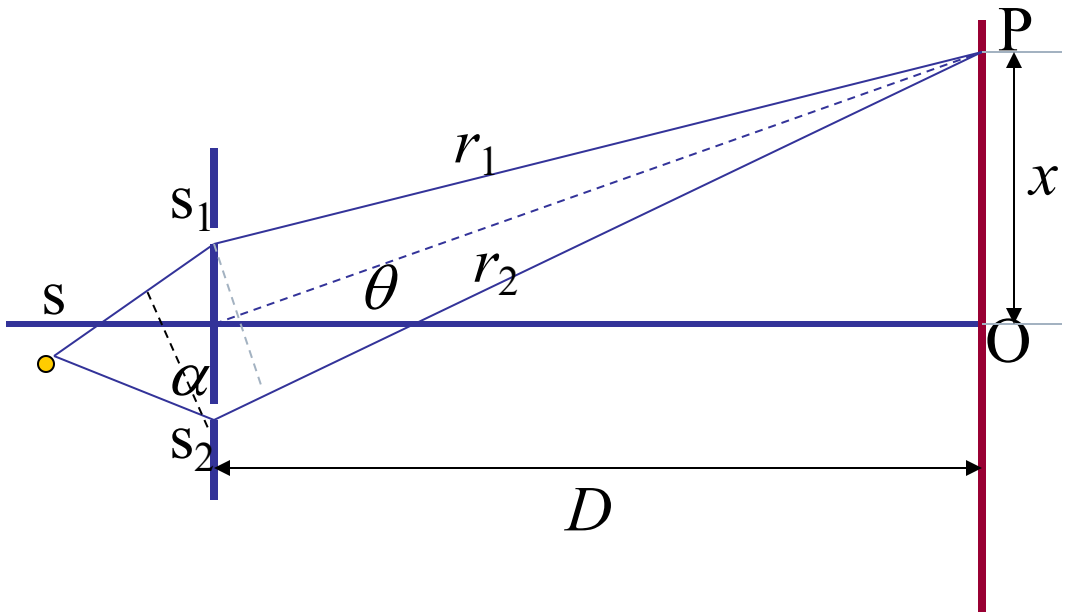
向上移动光源时，则：

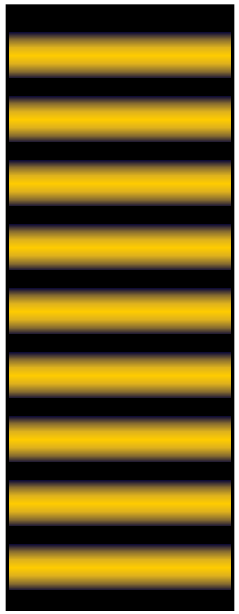


中央明纹下移

- 上下移动光源时，观察条纹的移动

向下移动光源时，则：





中央明纹上移

$$d\frac{x}{d'}+\frac{\lambda}{2}=\begin{cases}\pm k\lambda \\ \pm(2k+1)\dfrac{\lambda}{2}\end{cases} \qquad k=0,1,2,\cdots$$

例2 用很薄的云母片（n=1.58）覆盖在双缝实验中的一条缝上，这时零级明条纹移到原来的第七级明条纹位置上，如果入射光波长为550nm，求云母片厚度.

解: 覆盖云母片之前的零级明纹位置处光程差

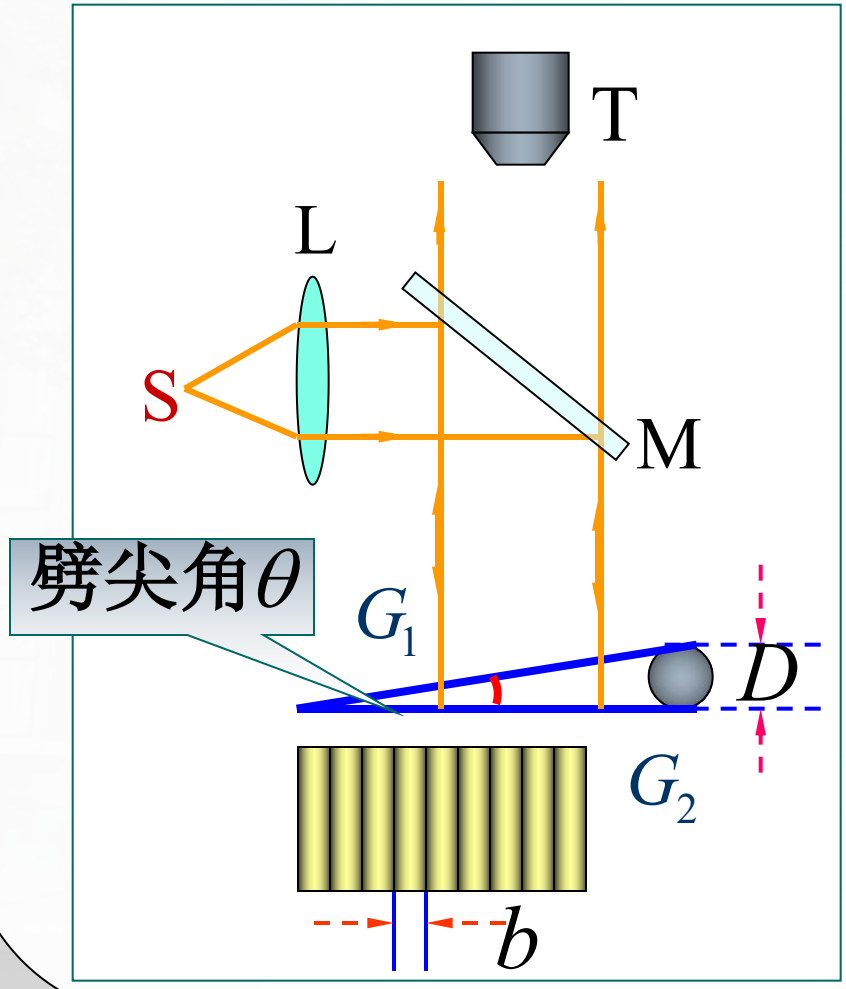
$$r_1 - r_2 = 0$$

设云母片厚度为e，覆盖云母片后x=0处的光程差

$$(r_1 - e) + ne - r_2 = 7\lambda$$

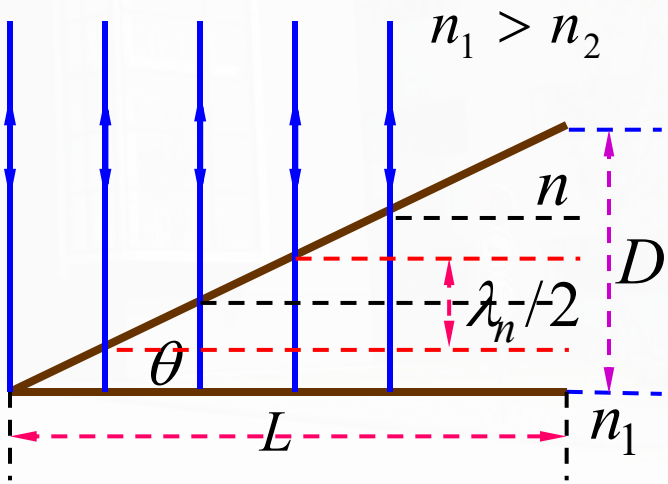
$$e = \frac{7\lambda}{n-1} = 6.64\times10^{-3}\,\text{mm}$$

二、劈尖



$$\Delta = 2n_2 d + \boxed{\frac{\lambda}{2}} \Leftarrow \boxed{\because n_2 < n_1}$$

$$\Delta = \begin{cases} k\lambda, & k = 1,2,\cdots \quad \text{明} \\ (2k+1)\dfrac{\lambda}{2}, & k = 0,1,\cdots \quad \text{暗} \end{cases}$$

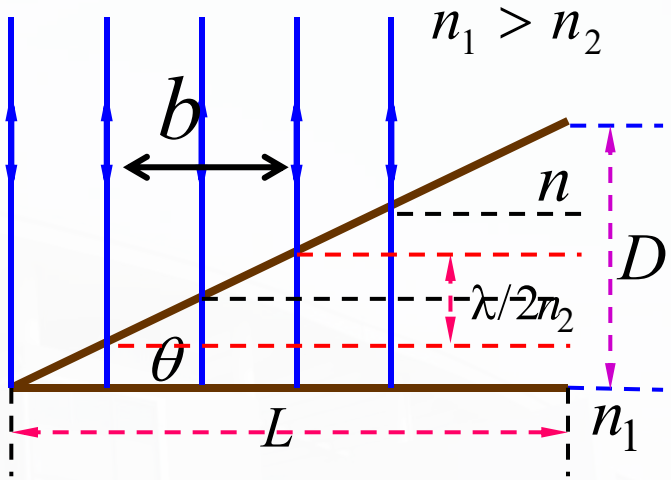


相邻明纹（暗纹）间的厚度差

$$h_{k+1} - h_k = \frac{\lambda}{2n_2}$$

λ光在真空中的波长

$$n_2 = 1, h_{k+1} - h_k = \frac{\lambda}{2}$$



$$D = \frac{\lambda}{2n_2 b} L$$

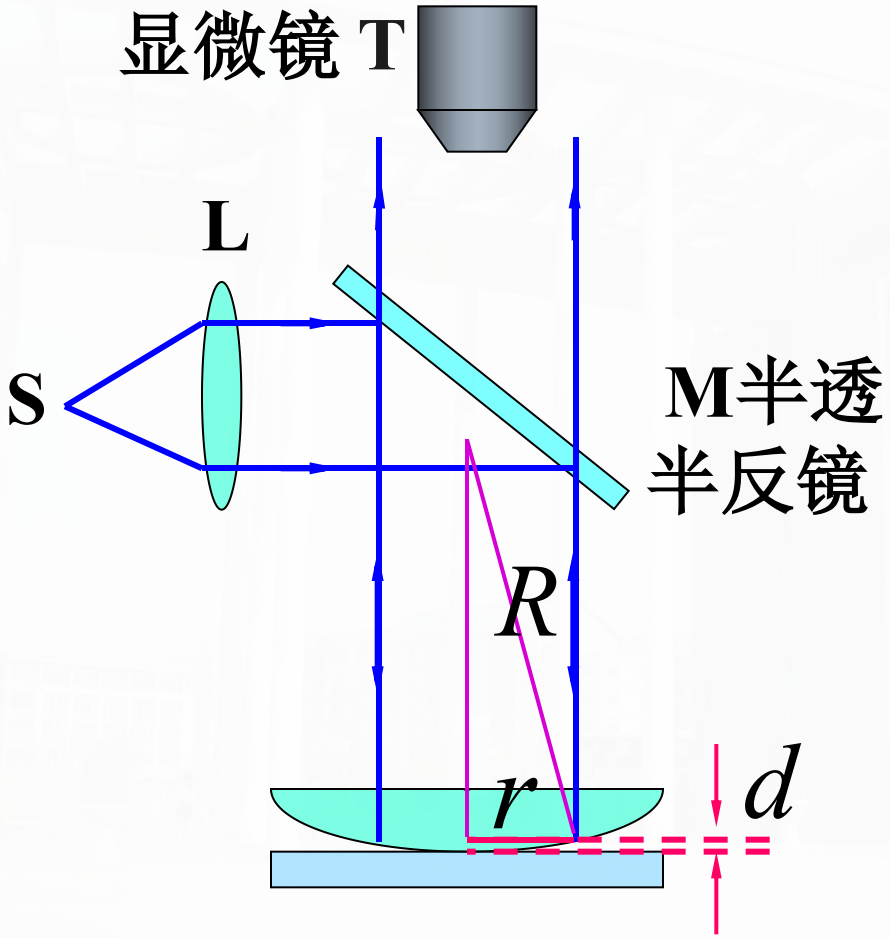
条纹间距（明纹或暗纹）

$$b\sin\theta = \frac{\lambda}{2n_2}, \sin\theta \approx \theta$$

$$\boxed{b = \frac{\lambda}{2n_2\theta}}$$

细丝直径

三、牛顿环



干涉图样

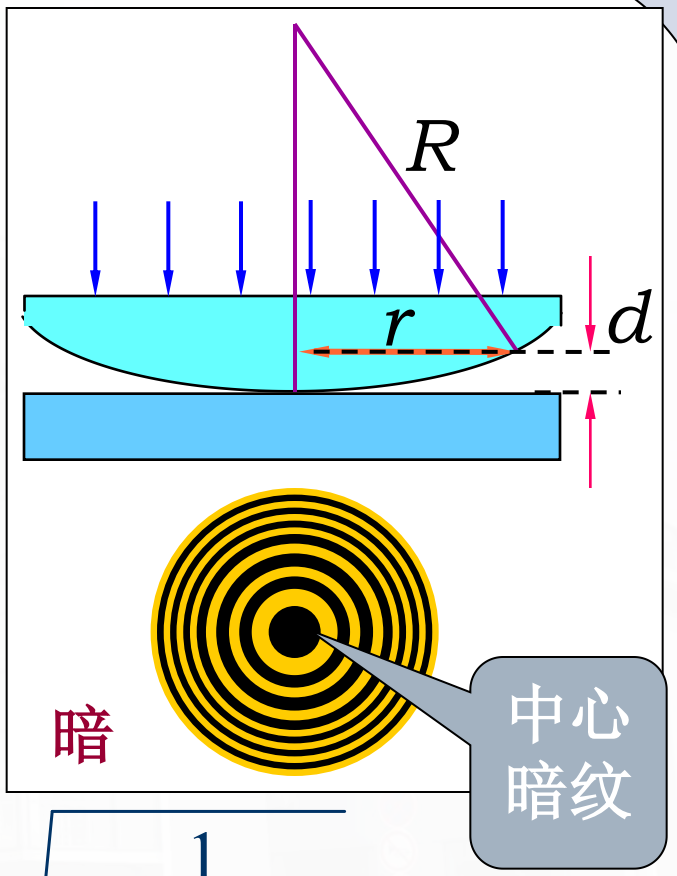
由图知

$$r^2 = R^2 - (R-d)^2 = 2dR - d^2$$

$$\because R >> d \quad \therefore d^2 \approx 0$$

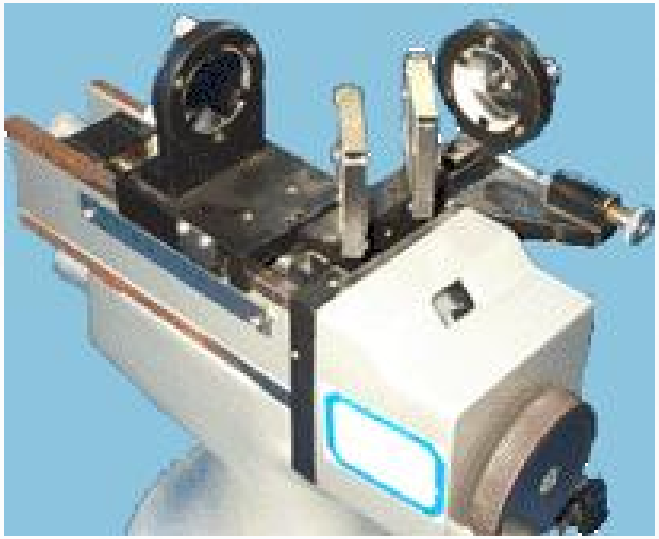
$$r = \sqrt{2dR}$$

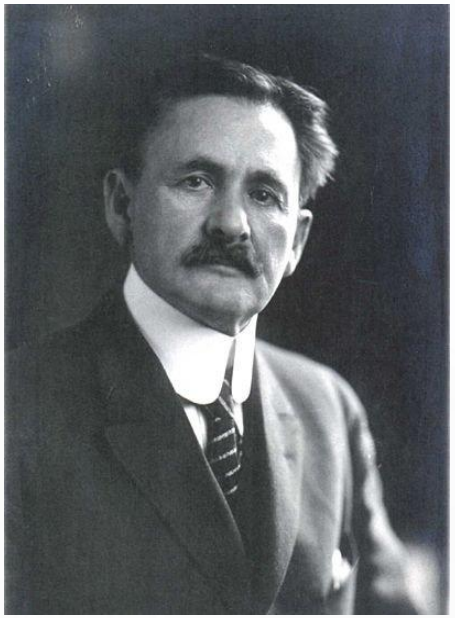
$$\Delta = 2d + \frac{\lambda}{2} = \begin{cases} k\lambda, & k = 1,2,\cdots \quad \text{明} \\ (2k+1)\dfrac{\lambda}{2}, & k = 0,1,\cdots \quad \text{暗} \end{cases}$$



$$r = \sqrt{2dR} = \sqrt{(\Delta - \frac{\lambda}{2})R} \Longrightarrow \begin{cases} r = \sqrt{(k - \frac{1}{2})R\lambda} & \text{明环半径} \\ r = \sqrt{kR\lambda} & \text{暗环半径} \end{cases}$$

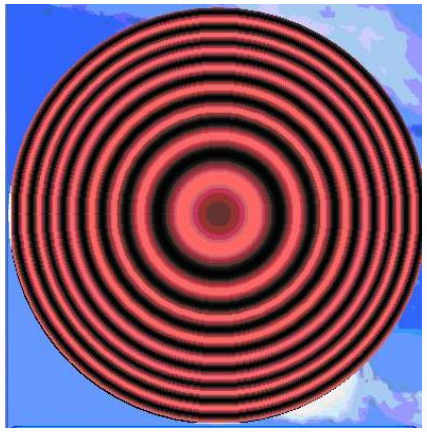
19－5 迈克耳逊干涉仪*

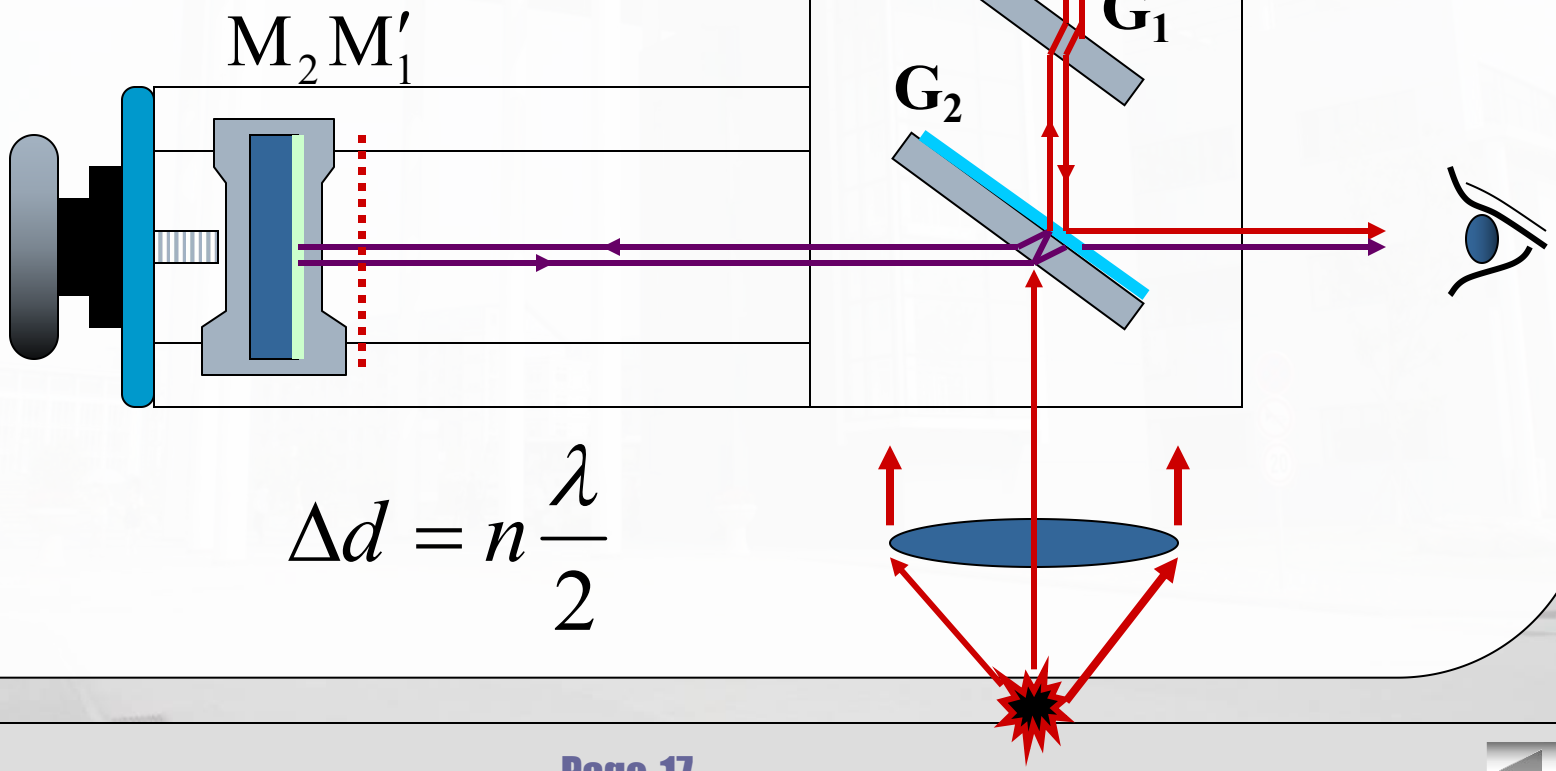




迈克耳逊干涉仪(Michelson interferometer)是利用光的干涉精确测量长度和长度变化的仪器。

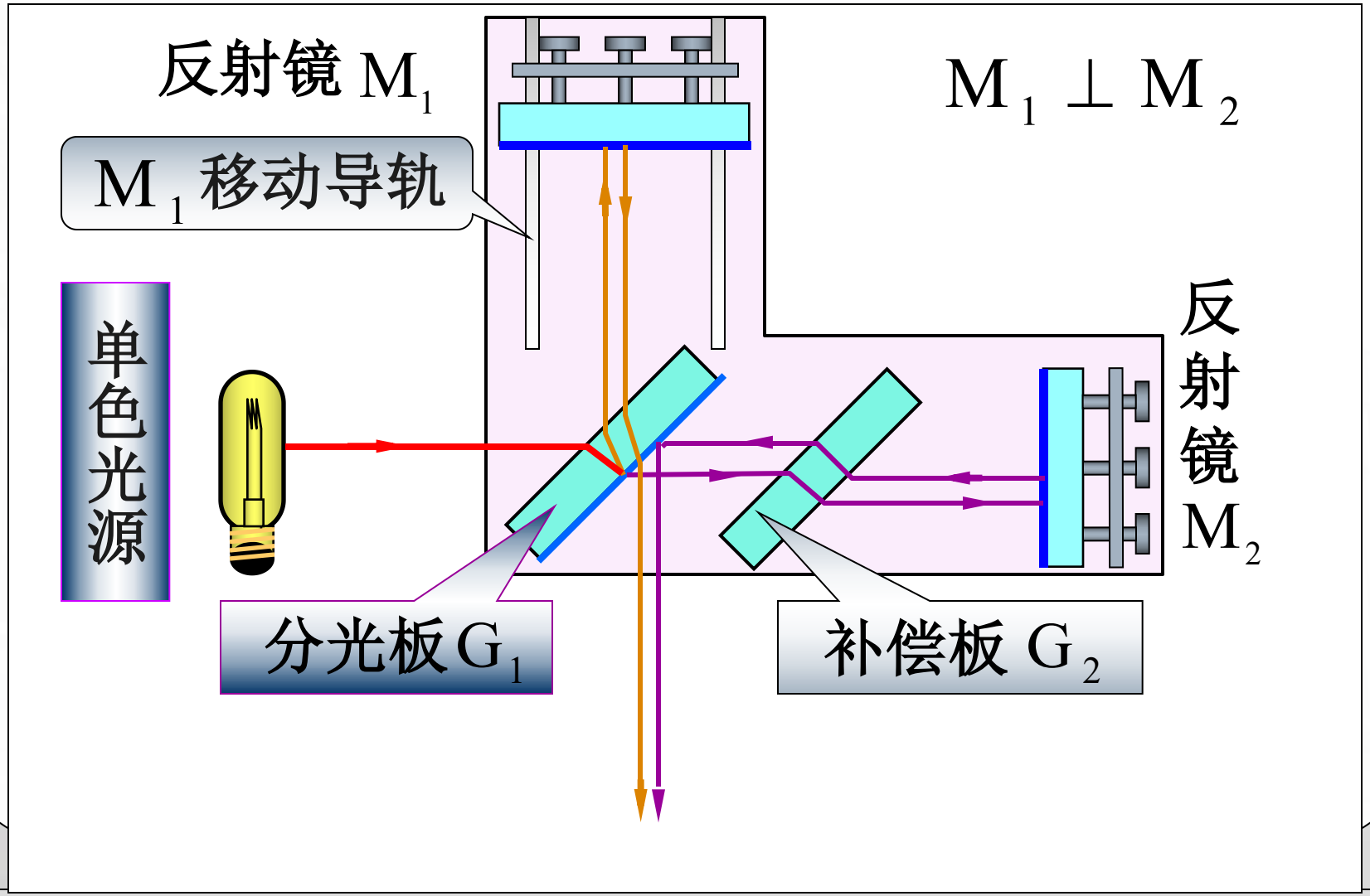
迈克耳逊和莫雷曾利用它进行过著名的否定旧以太说的实验，在物理学中占有一定地位。

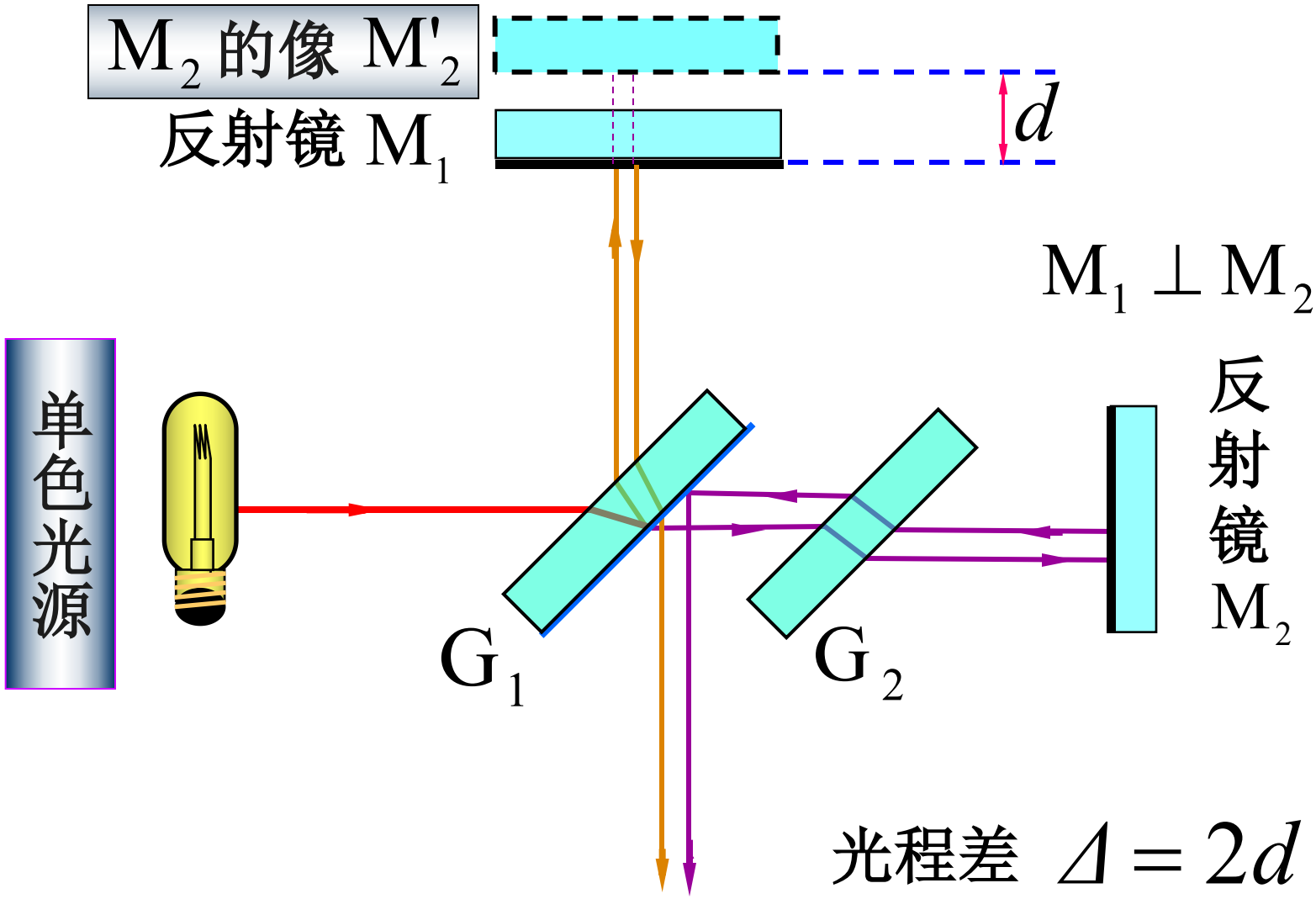




$$\Delta d = n\frac{\lambda}{2}$$

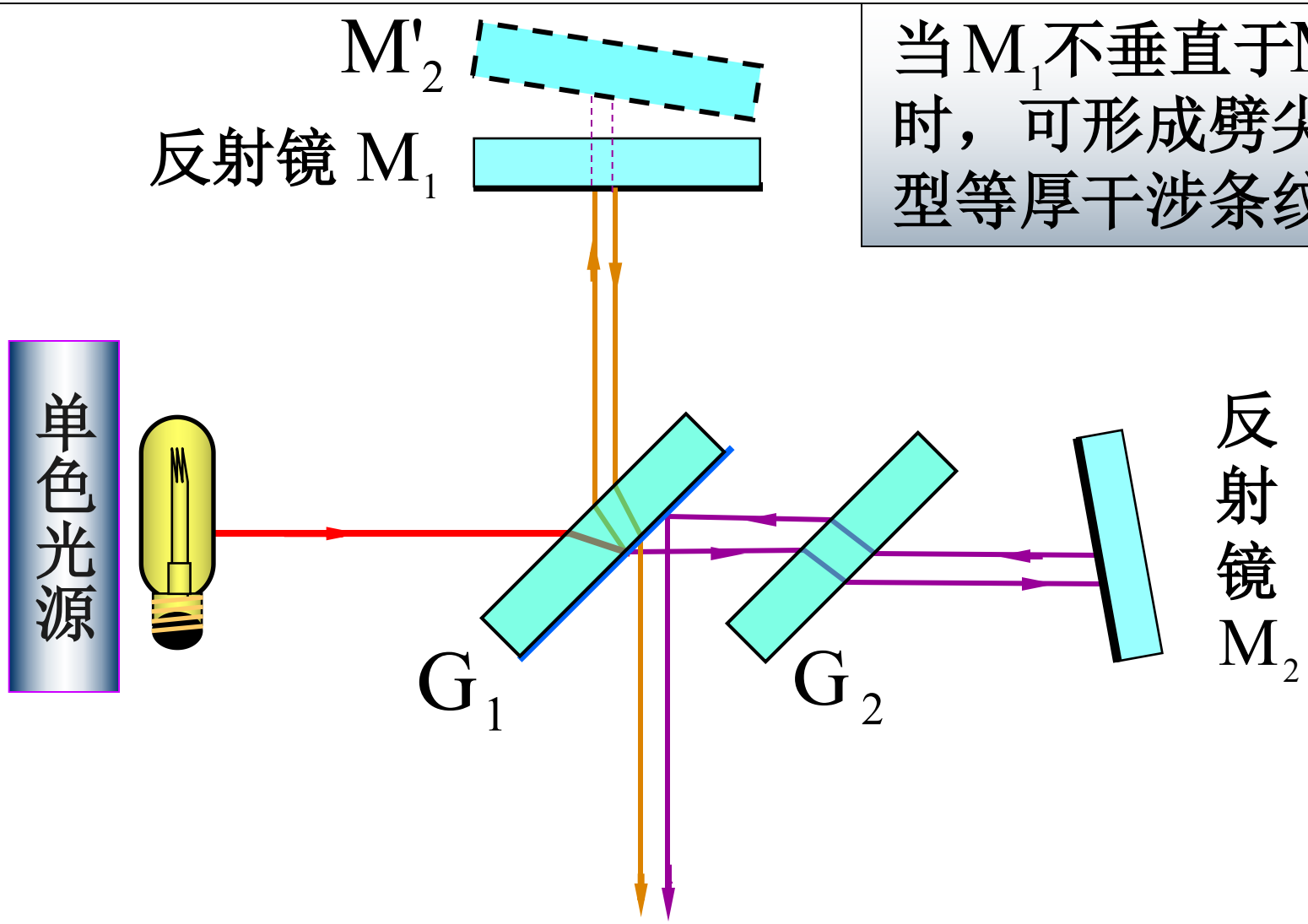
$G_1 // G_2$ 与 M_1, M_2 成 45^0角





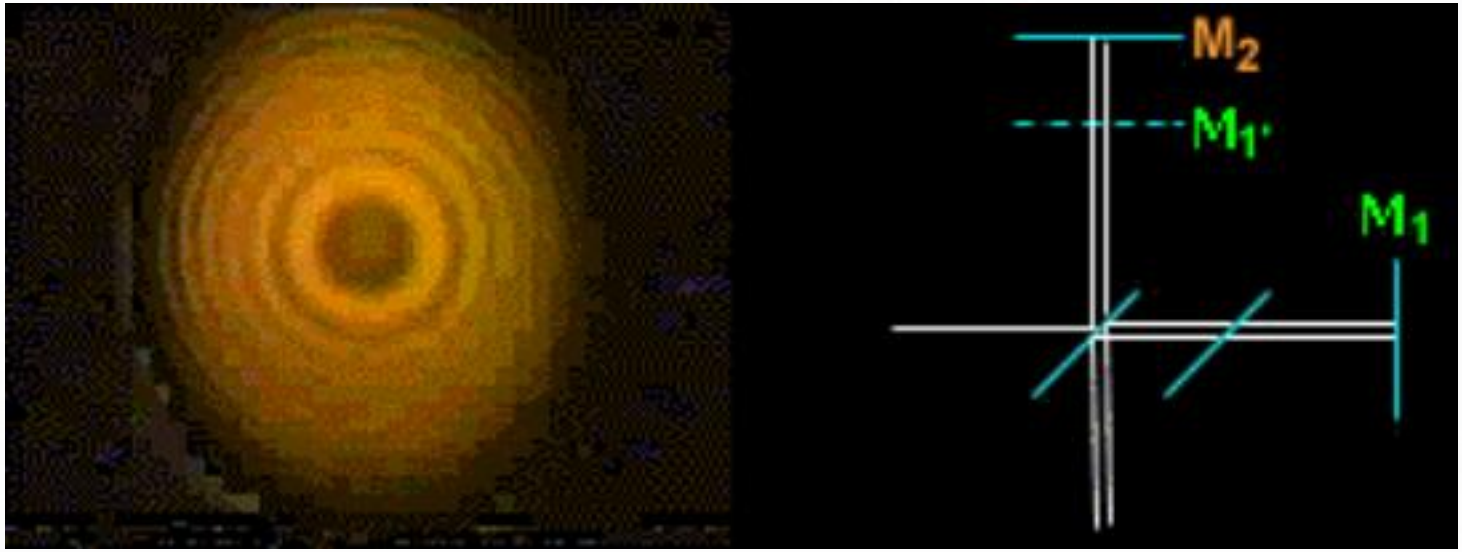
光程差 $\Delta = 2d$

当M_1不垂直于M_2时，可形成劈尖型等厚干涉条纹.

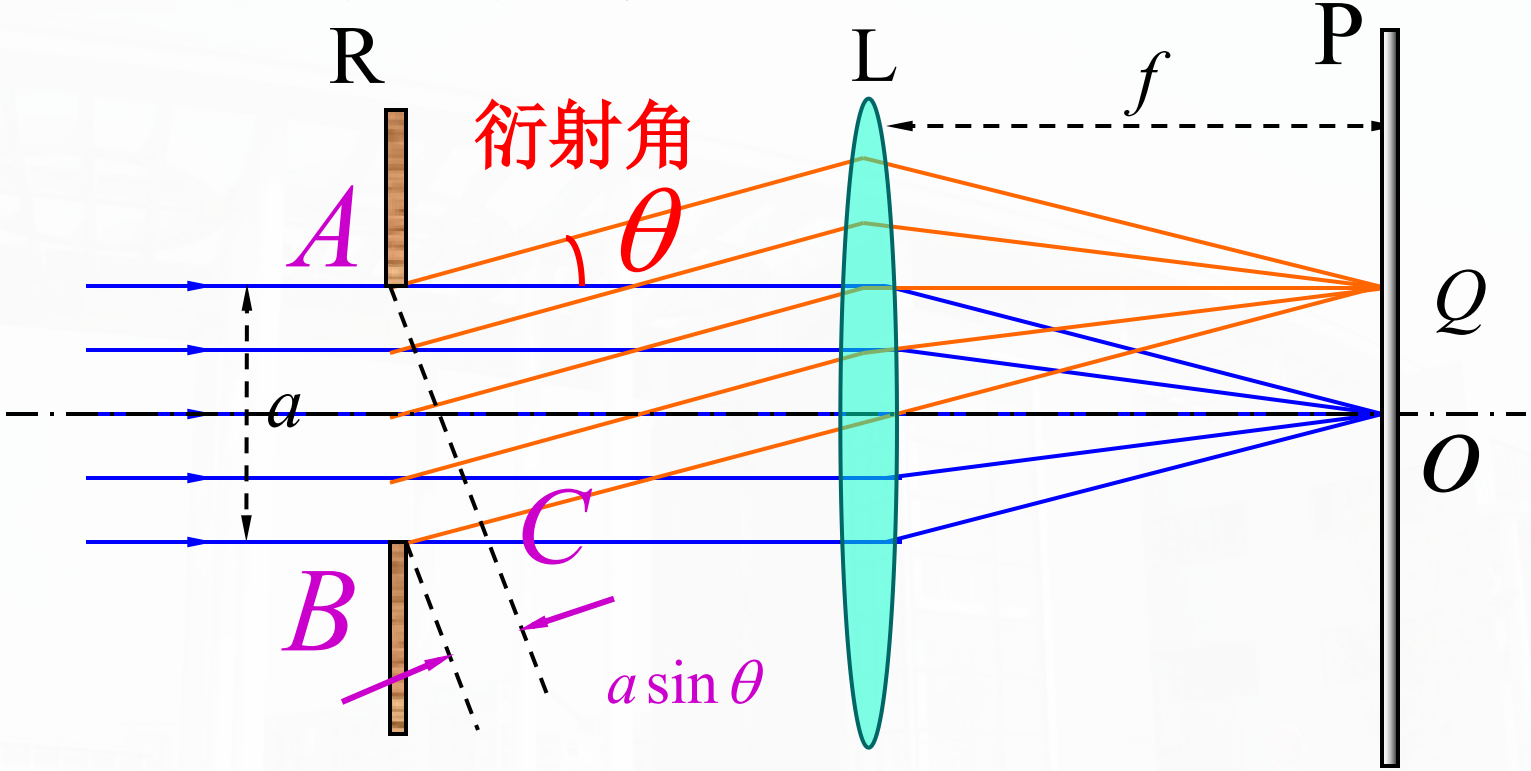


干涉条纹的移动

当 M_1 与 M'_2之间距离变大时，圆形干涉条纹从中心一个个长出，并向外扩张，干涉条纹变密；距离变小时，圆形干涉条纹一个个向中心缩进，干涉条纹变稀．



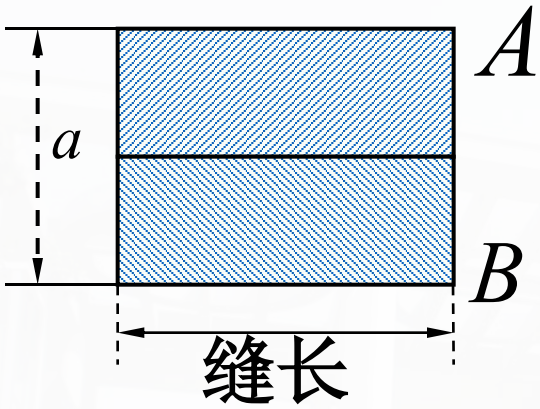
19－8 夫琅禾费单缝衍射



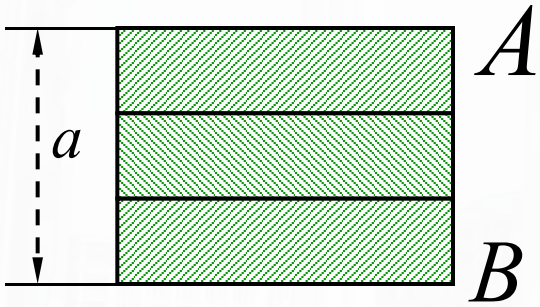
（衍射角 θ：向上为正，向下为负 .）

菲涅尔波带法

$$BC = a\sin\theta = \pm k\frac{\lambda}{2} \quad (k = 1,2,3,\cdots)$$

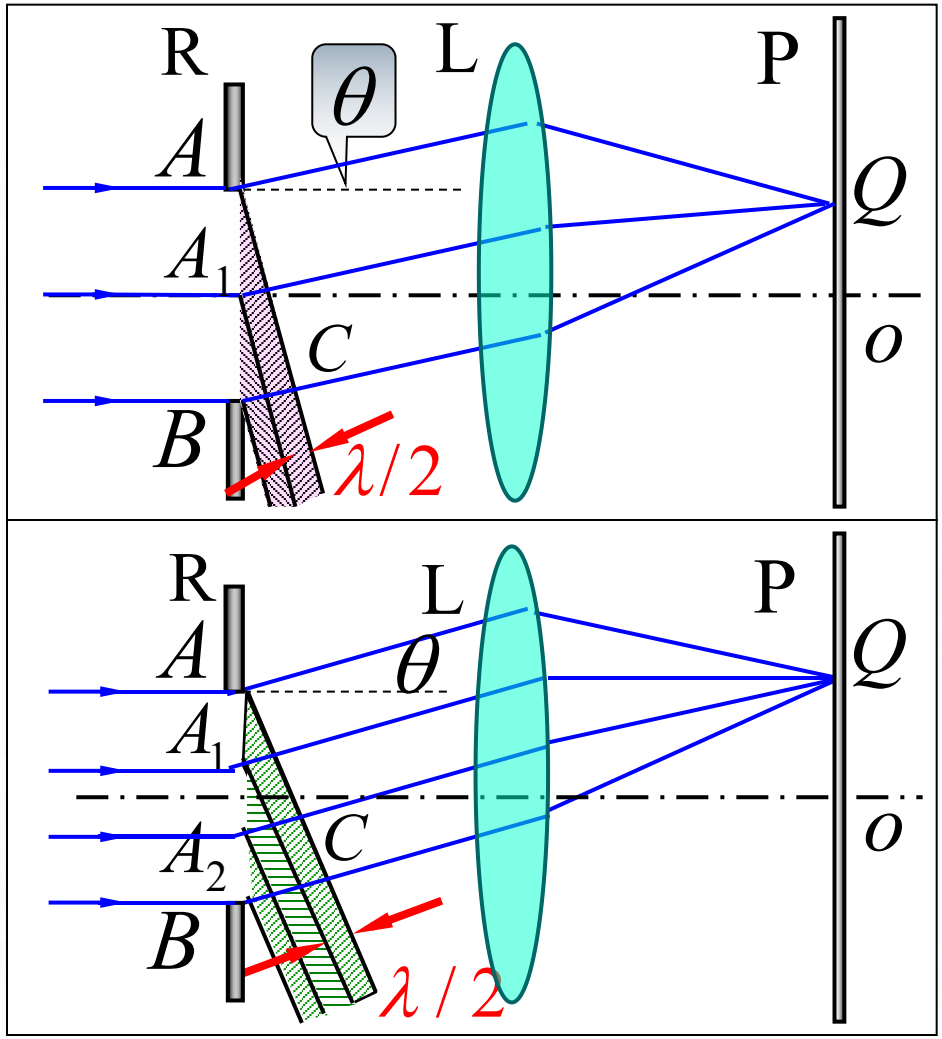


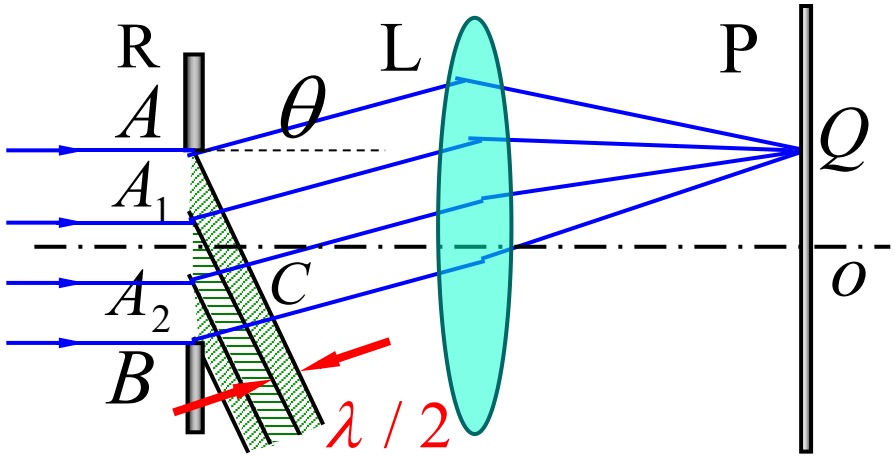
$$a\sin\theta = \pm 2k\,\lambda/2$$



$$a\sin\theta = \pm(2k+1)\,\lambda/2$$

$$k = 1, 2, 3, \cdots$$





$$BC = a\sin\theta = \pm k\frac{\lambda}{2}$$

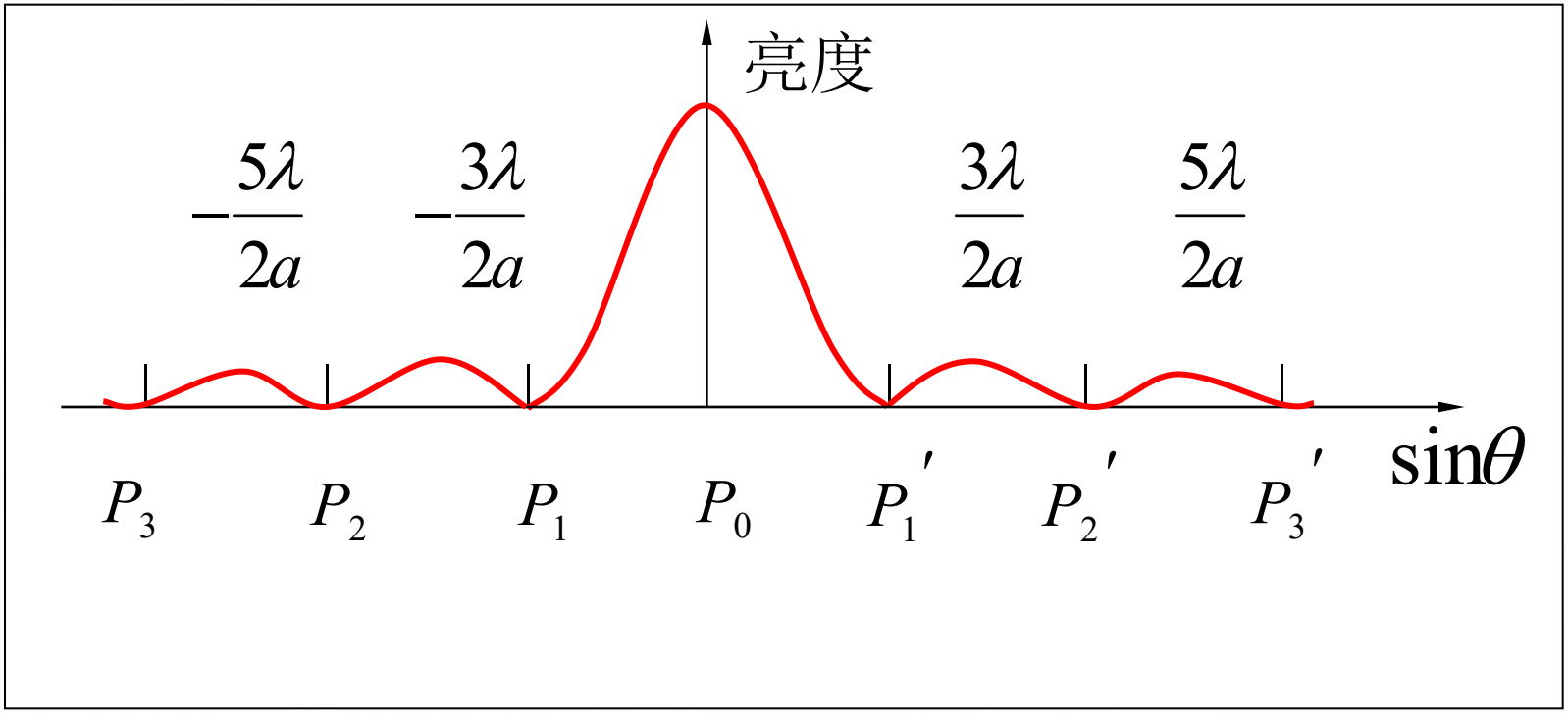
（k 个半波带）

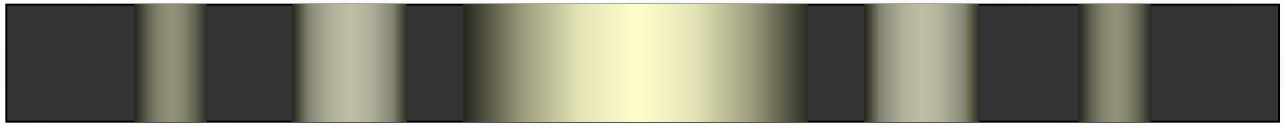
$$a\sin\theta = 0$$ 中央明纹中心

$$a\sin\theta = \pm 2k\frac{\lambda}{2} = \pm k\lambda$$ 干涉相消（暗纹） $2k$ 个半波带

$$a\sin\theta = \pm(2k+1)\frac{\lambda}{2}$$ 干涉加强（明纹） $2k+1$ 个半波带

$$a\sin\theta \neq k\frac{\lambda}{2}$$ （介于明暗之间） $(k = 1, 2, 3, \cdots)$


亮度
$-\frac{5\lambda}{2a}$
$-\frac{3\lambda}{2a}$
$\frac{3\lambda}{2a}$
$\frac{5\lambda}{2a}$
P_3
P_2
P_1
P_0
P_1'
P_2'
P_3'
$\sin\theta$



讨 论

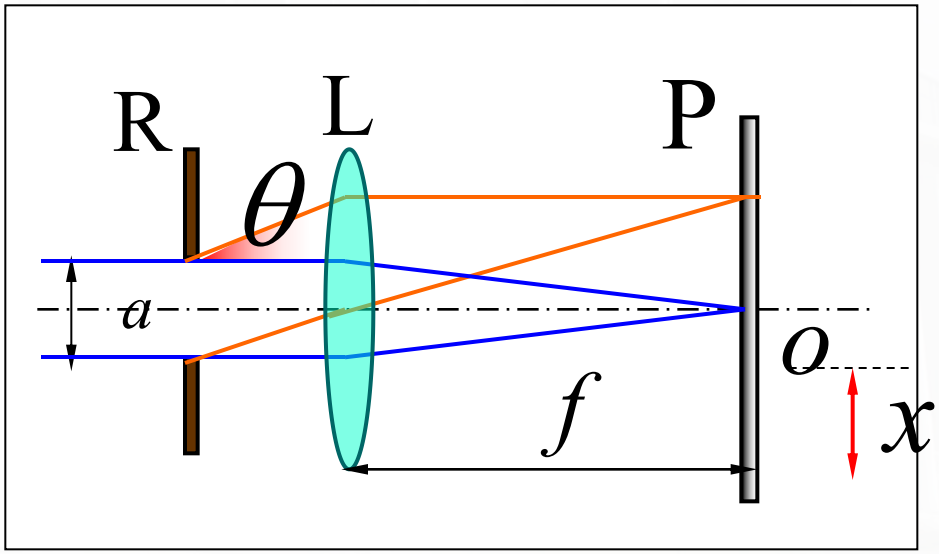
$$\sin\theta \approx \theta, \quad x = \theta f, \quad a\sin\theta \approx a\frac{x}{f}$$

（1）第一暗纹距中心的距离

$$x_1 = \theta f = \frac{\lambda}{a} f$$

第一暗纹的衍射角

$$\theta_1 = \arcsin\frac{\lambda}{a}$$



第一暗纹的衍射角

$$\theta_1 = \arcsin\frac{\lambda}{a}$$

1. λ 一定

a 增大，θ_1 减小 $\frac{\lambda}{a} \Rightarrow 0, \theta_1 \Rightarrow 0$ 光直线传播

a 减小，θ_1 增大 $a \Rightarrow \lambda, \theta_1 \Rightarrow \frac{\pi}{2}$ 衍射最大

2. a 一定，λ 越大，θ_1 越大，衍射效应越明显.

（2）中央明纹

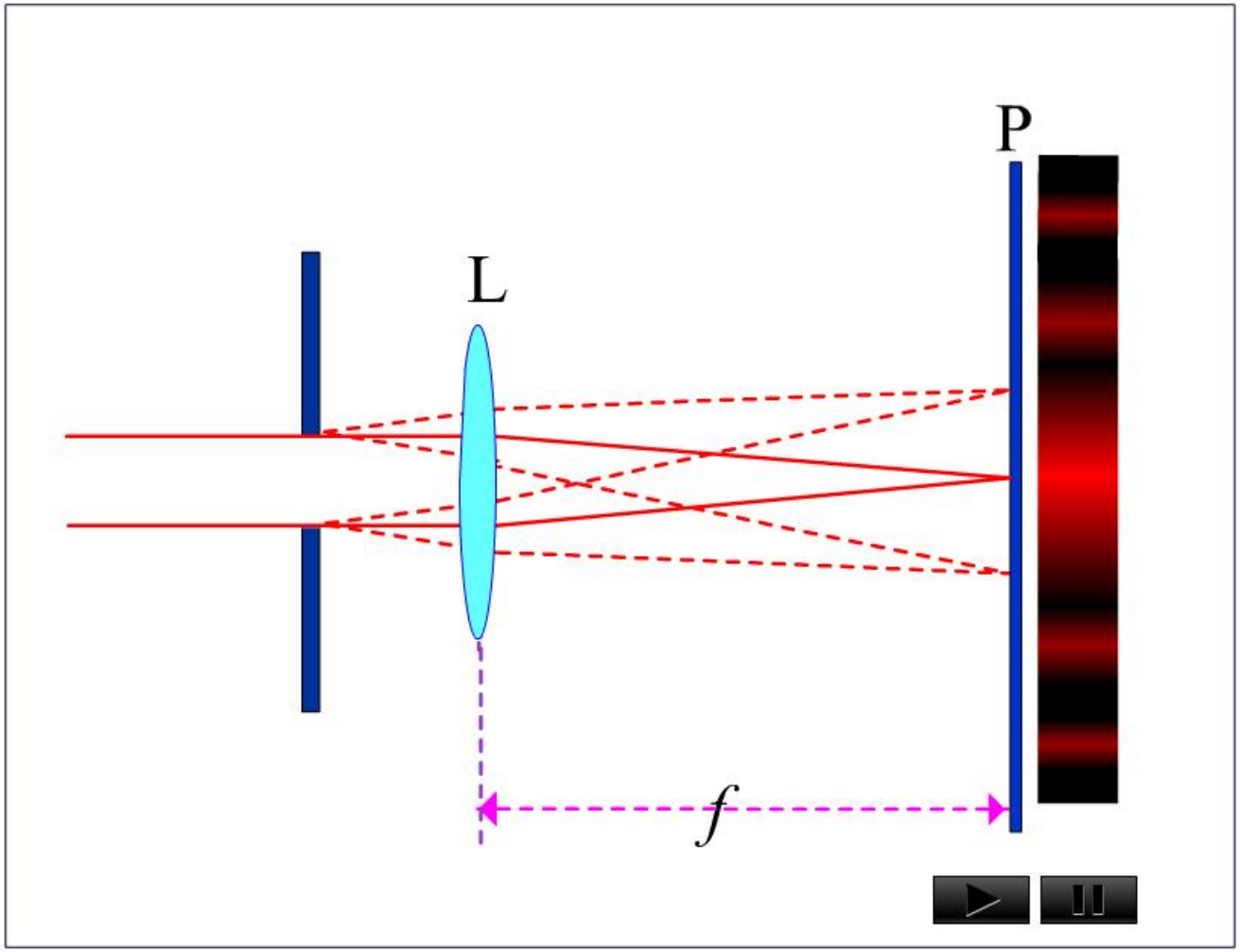
（$k=1$的两暗纹间）

角范围 $-\frac{\lambda}{a}<\sin\theta<\frac{\lambda}{a}$ 线范围 $-\frac{\lambda}{a}f<x<\frac{\lambda}{a}f$

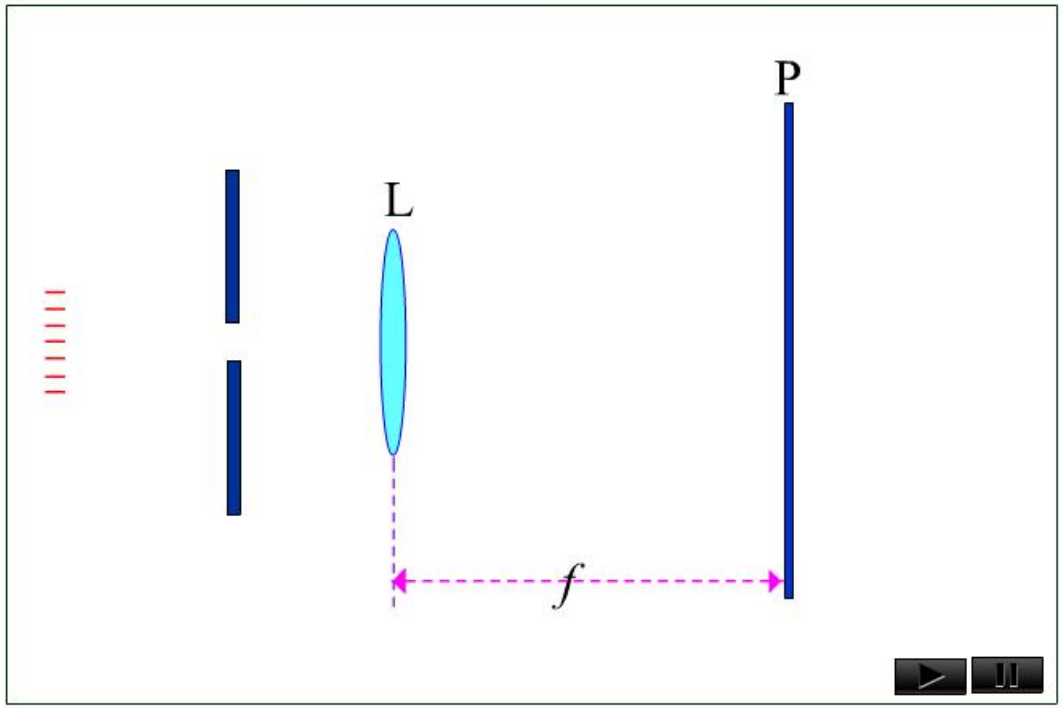
中央明纹的宽度

$$l_0=2x_1\approx 2\frac{\lambda}{a}f$$

单缝宽度变化



入射波长变化



λ越大，θ_1越大，衍射效应越明显.

（3）条纹宽度（相邻条纹间距）

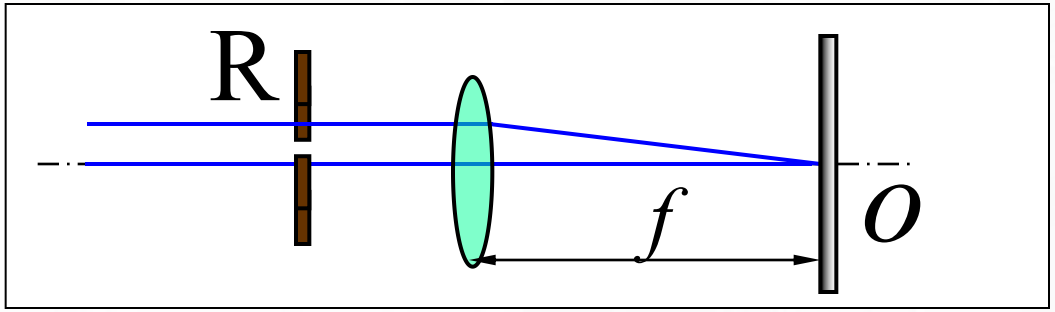
$$\begin{cases} a\sin\theta = \pm 2k\dfrac{\lambda}{2} = \pm k\lambda & \text{干涉相消（暗纹）} \\ a\sin\theta = \pm(2k+1)\dfrac{\lambda}{2} & \text{干涉加强（明纹）} \end{cases}$$

$$l = \theta_{k+1}f - \theta_k f = \frac{\lambda f}{a}$$

除了中央明纹外的其它明纹、暗纹的宽度

（4）单缝衍射的动态变化

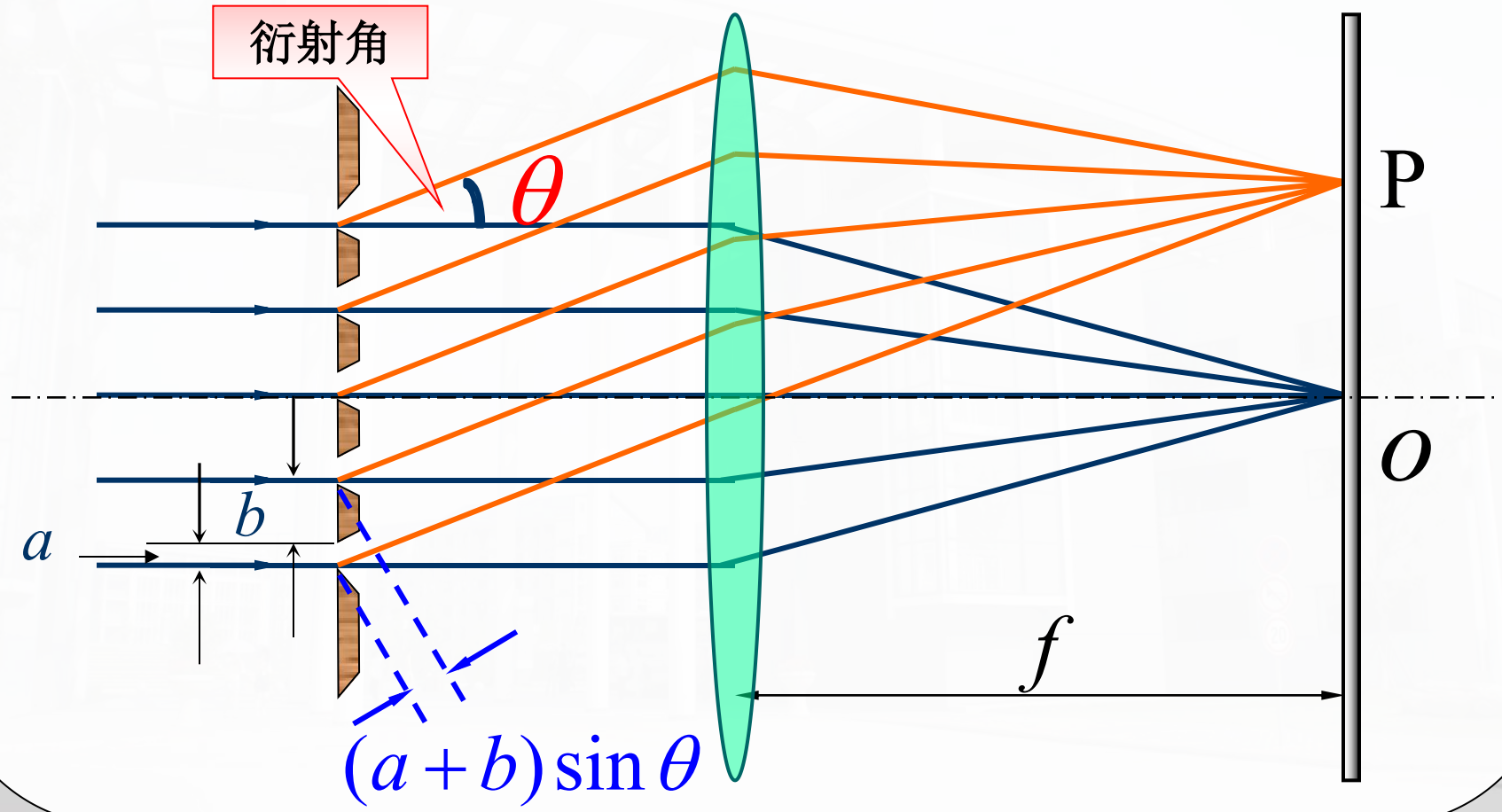
单缝上下移动，根据透镜成像原理衍射图不变 .



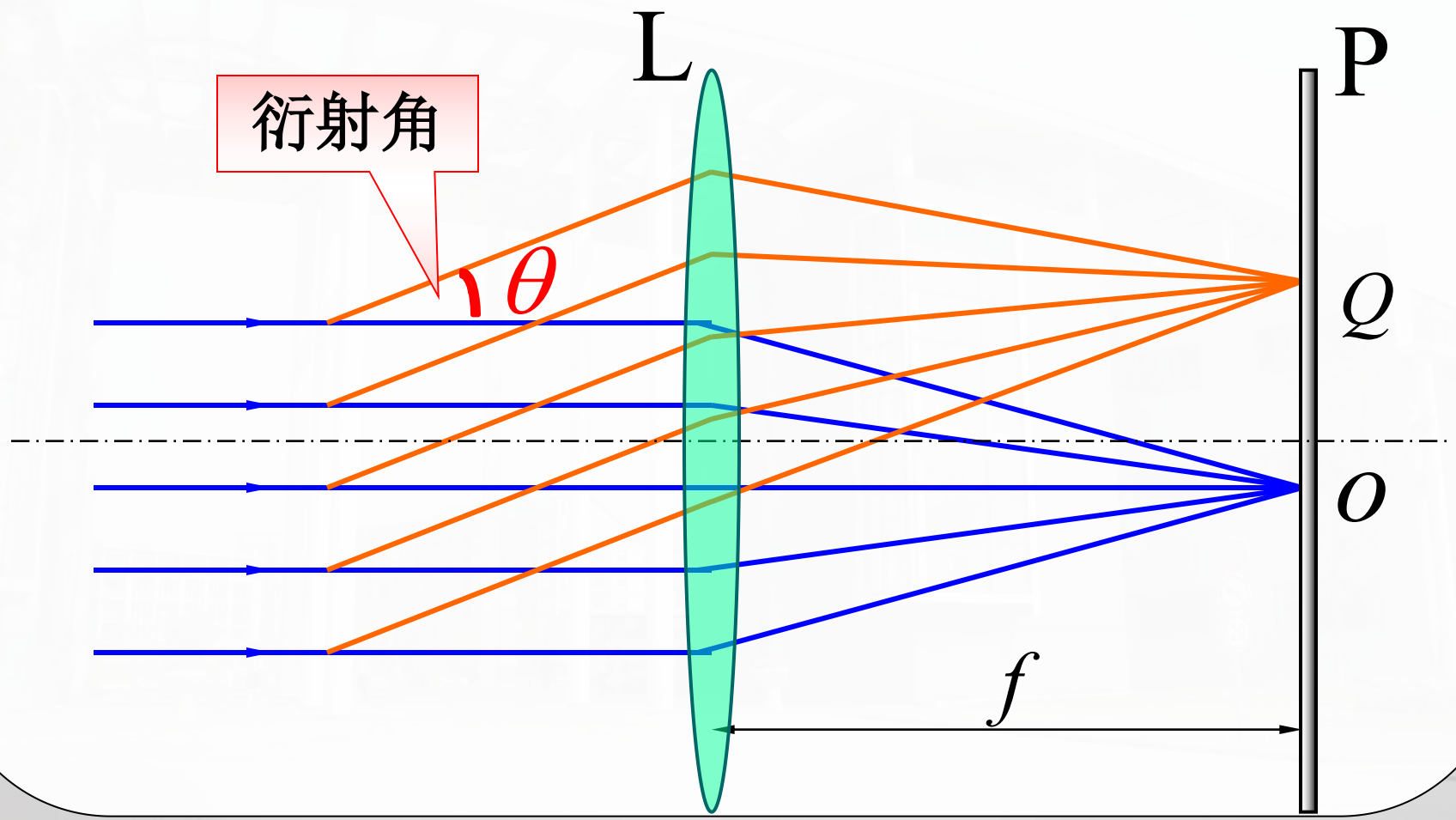
单缝上移，零级明纹仍在透镜光轴上.

19－10 光栅衍射及光栅光谱

一、光栅衍射



光栅：许多等宽度、等距离的狭缝排列起来形成的光学元件.



1. 光栅方程

光栅的衍射条纹是衍射和干涉的总效果

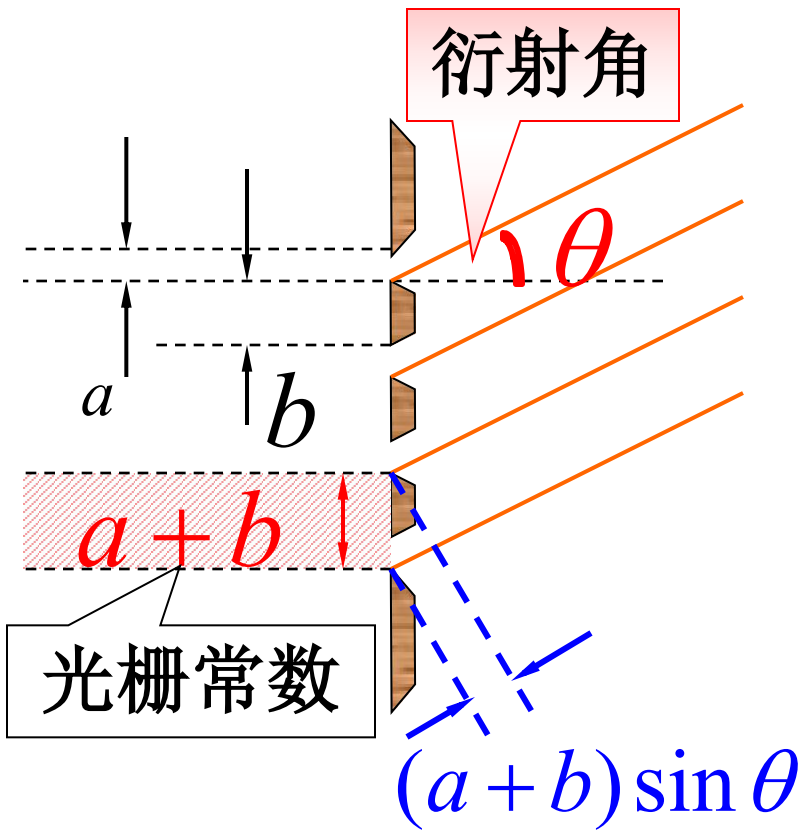
相邻两缝间的光程差：

$$\Delta = (a+b)\sin\theta$$

明纹位置

$$(a+b)\sin\theta = \pm k\lambda$$

$$(k = 0,1,2,\cdots)$$



a ：透光部分的宽度

b ：不透光部分的宽度

光栅常数：$10^{-5} \sim 10^{-6}\,\mathrm{m}$

2. 主极大

满足光栅方程的明纹又称主明纹或主极大条纹，也称光谱线。

$k=0$时，$\theta=0$ 称中央明条纹；$k=\pm1,\pm2...$ 分别称为第一级、第二级主极大，正负号表示各级明纹对称地分布在中央明纹两侧。

说明：

主极大位置由缝间干涉决定；

主极大的最大级数为 $k<\dfrac{(a+b)}{\lambda}$ 因为 $|\sin\theta|\le1$

3. 缺级

明纹极大由光栅各缝的干涉在屏上形成，但由于单缝衍射的影响，当暗纹衍射角对应干涉极大时，虽然满足光栅，但对应衍射角 θ 的主极大并不出现，这称为光谱线的缺级，缺级的级数 k 为

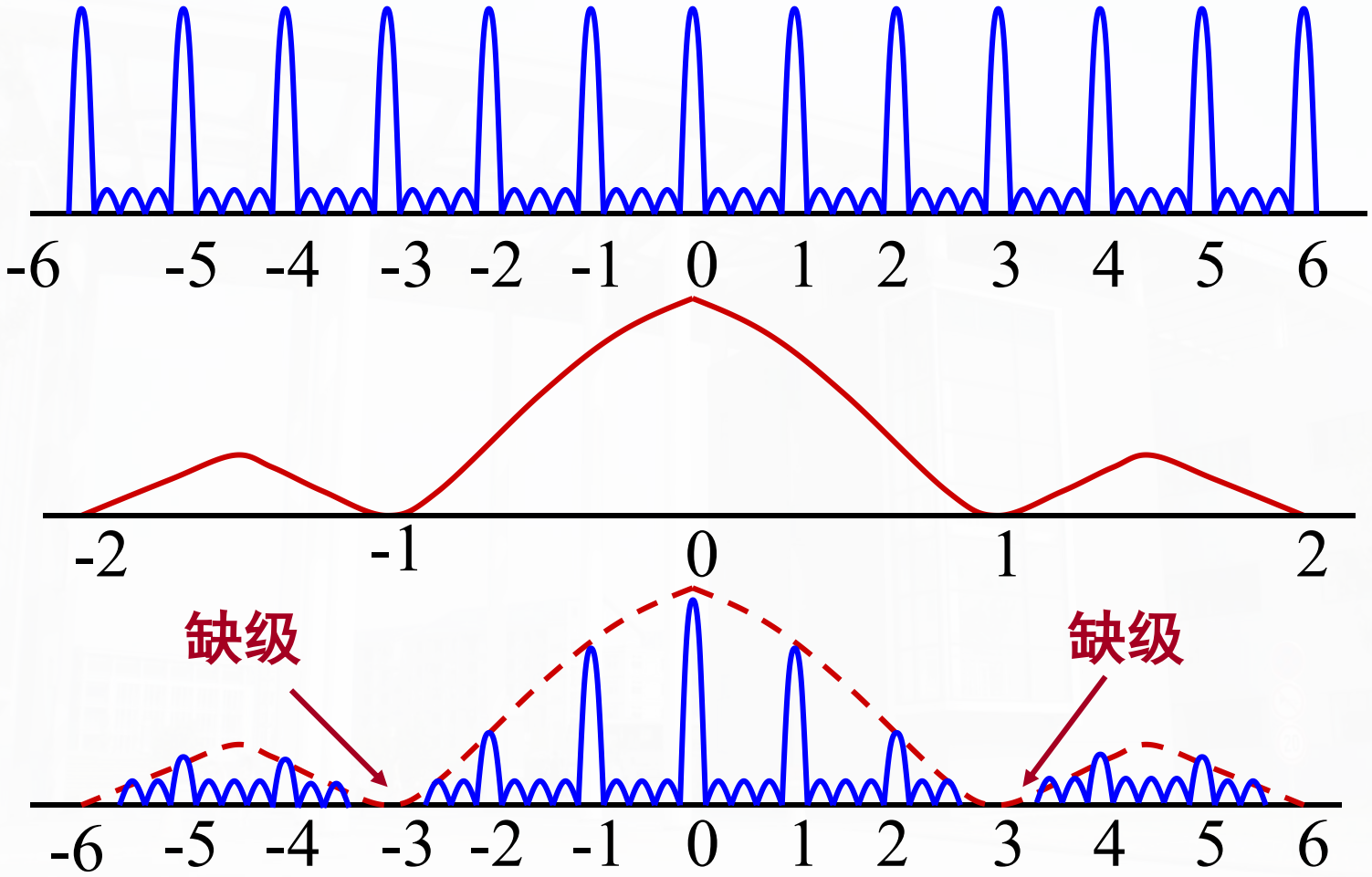
单缝衍射 $$a\sin\theta = k'\lambda$$

光栅方程 $$(a+b)\sin\theta = k\lambda$$

同时满足二式时，光谱线k级将不出现

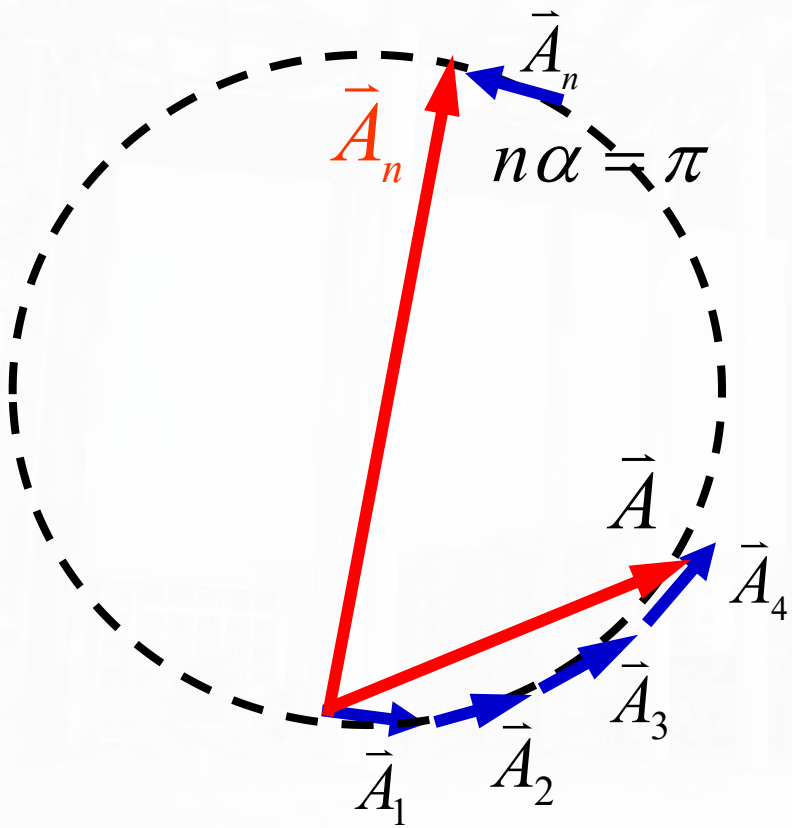
$$k = \frac{a+b}{a}k'$$

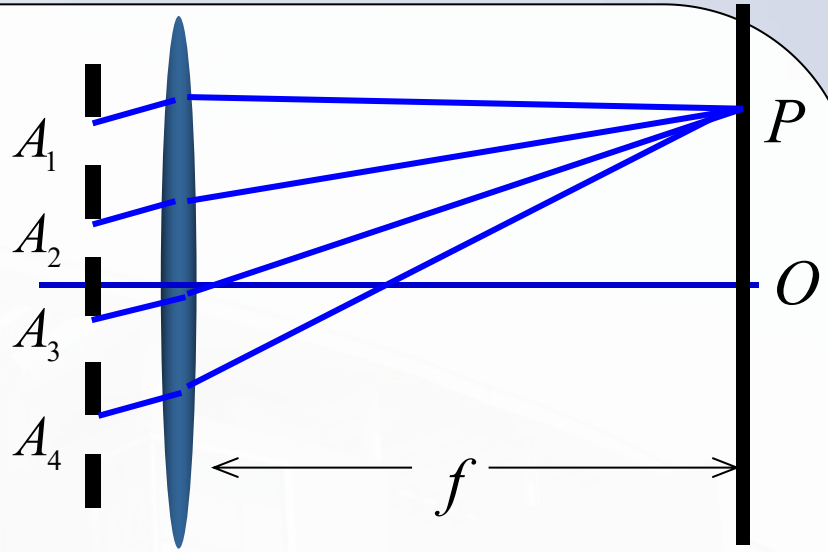
衍射对多缝干涉的影响（缺级现象）



4. 暗纹

光矢量叠加分析：





$$N\alpha = N\frac{2\pi}{\lambda}\Delta = m\cdot 2\pi$$

干涉相消

$$N\alpha = m\cdot 2\pi$$

$$N\Delta = m\lambda$$

（m=1 ,2 ,…）

$$N\Delta = N(a+b)\sin\varphi = m\lambda$$

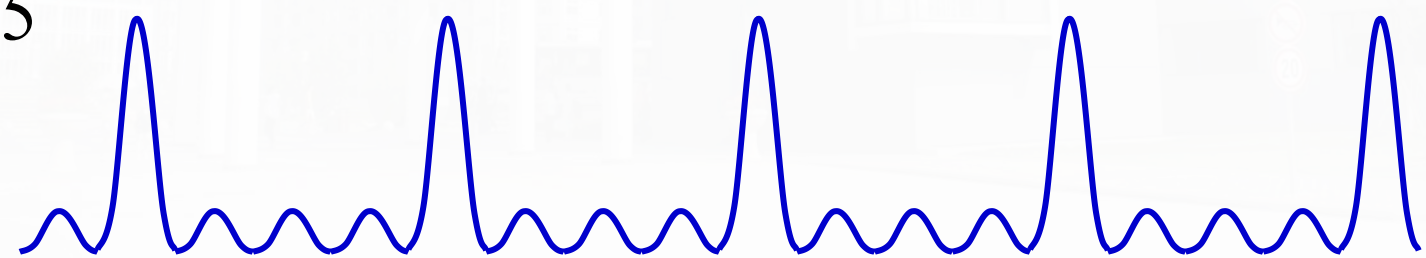
光栅暗纹 $$(a+b)\sin\varphi = \pm\frac{m}{N}\lambda$$

$$m = 1,2,\cdots,N-1,N+1,N+2,\cdots,2N-1,2N+1,\cdots$$

当： $m = N, 2N, 3N, \cdots \quad \Rightarrow k = \frac{m}{N} = 1, 2, \cdots$

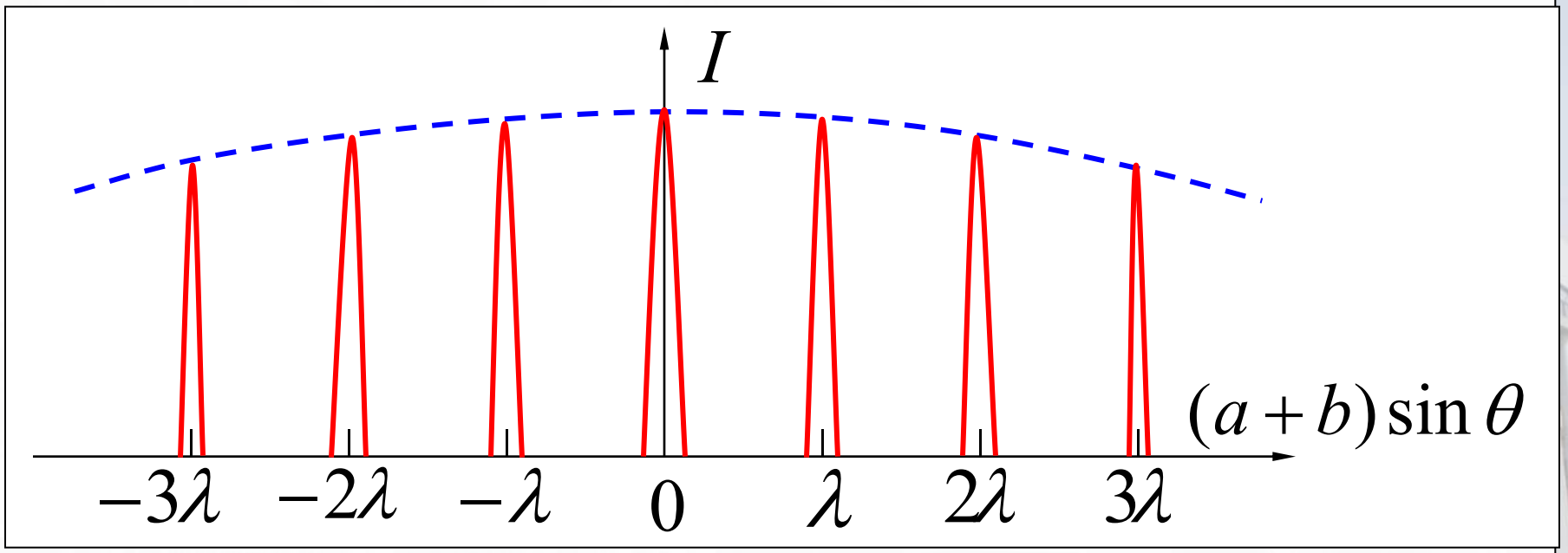
$$(a+b)\sin\varphi = \pm k\lambda$$

N=5



讨 论 $(a+b)\sin\theta = \pm k\lambda \quad (k = 0,1,2,\cdots)$

- 光强分布



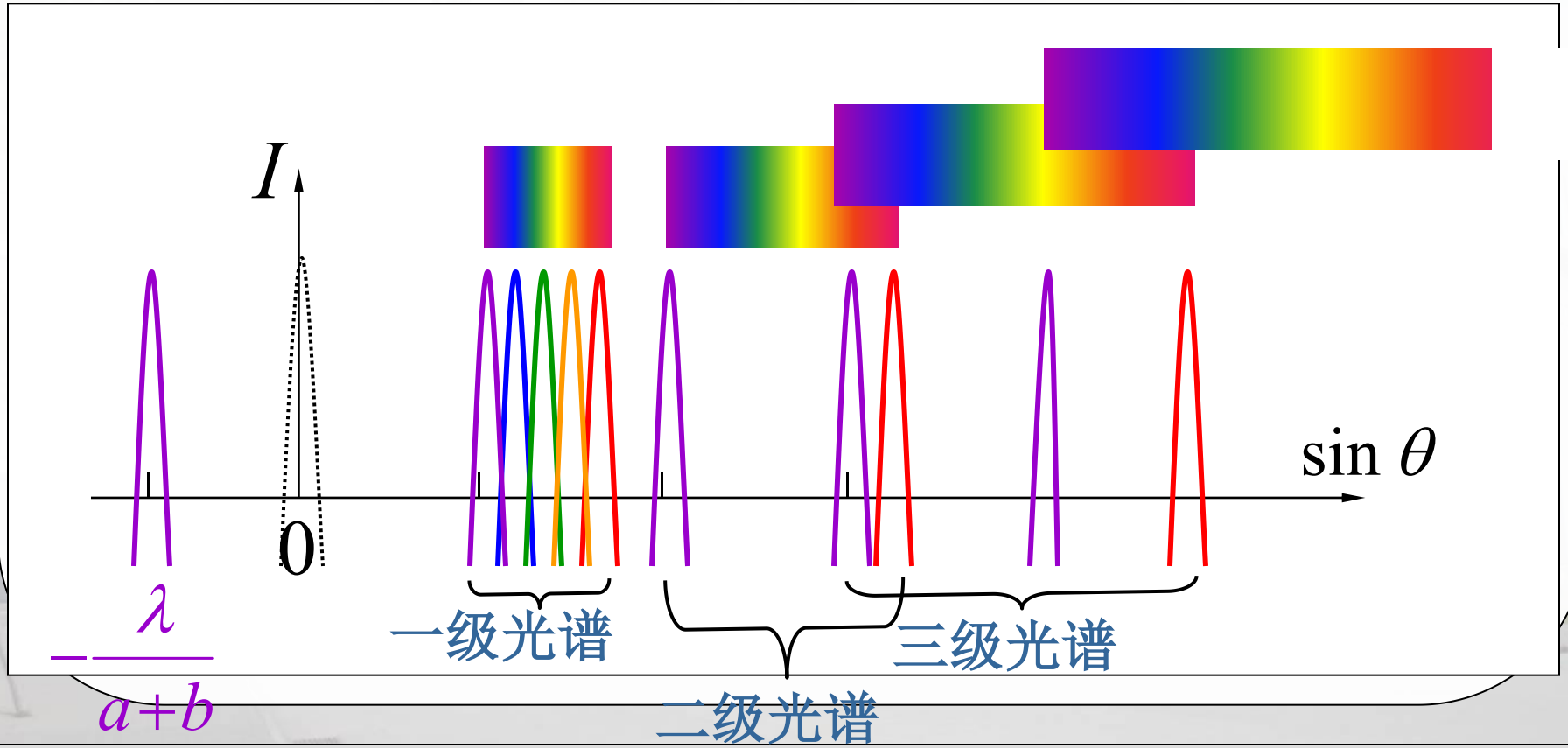
- 条纹最高级数

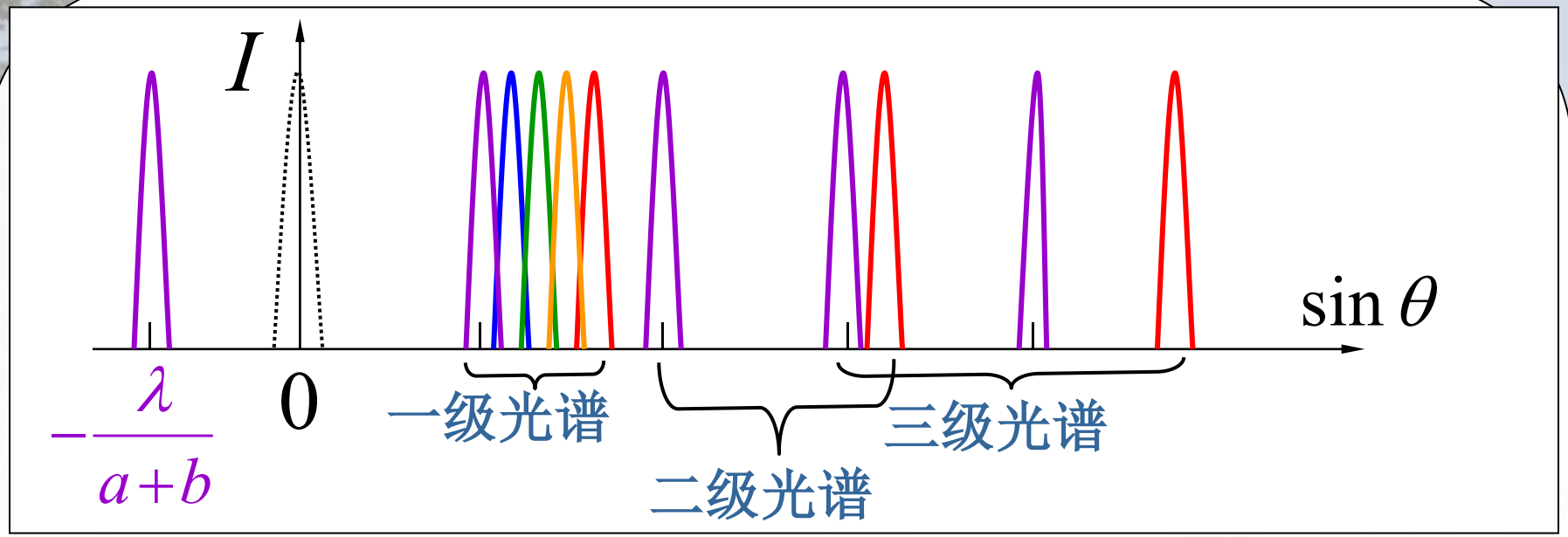
$$\sin\theta_k = \pm\frac{k\lambda}{a+b} \qquad \theta = \pm\frac{\pi}{2}, \qquad k = k_{\max} = \frac{a+b}{\lambda}$$

二、衍射光谱

$$(a+b)\sin\theta = \pm k\lambda \qquad (k = 0,1,2,\cdots)$$

入射光为白光时，λ不同，θ_k不同，按波长分开形成光谱.





例如 二级光谱重叠部分光谱范围

$$\begin{cases}(a+b)\sin\theta = 3\lambda_{紫} \\ (a+b)\sin\theta = 2\lambda\end{cases}$$

$$\lambda = 400 \sim 760\text{nm}$$

$$\lambda = \frac{3}{2}\lambda_{紫} = 600\text{nm}$$

二级光谱重叠部分：

$$600 \sim 760\text{nm}$$

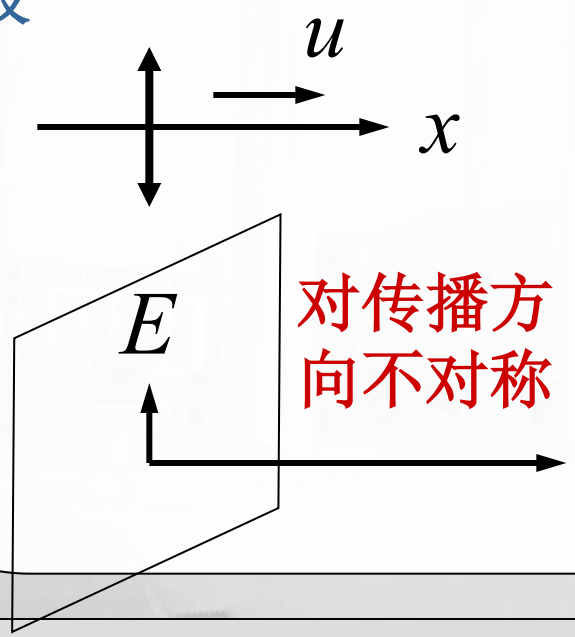
19－13 光的偏正性 马吕斯定律

光波：特定频率范围内的电磁波。

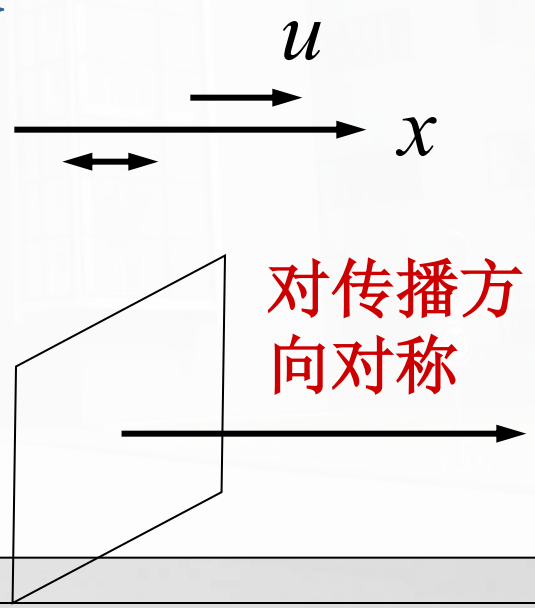
光矢量(light vector)：电磁波的电场强度 $\vec{E}$ 矢量。

电磁波是横波，光矢量振动方向与光传播方向垂直。

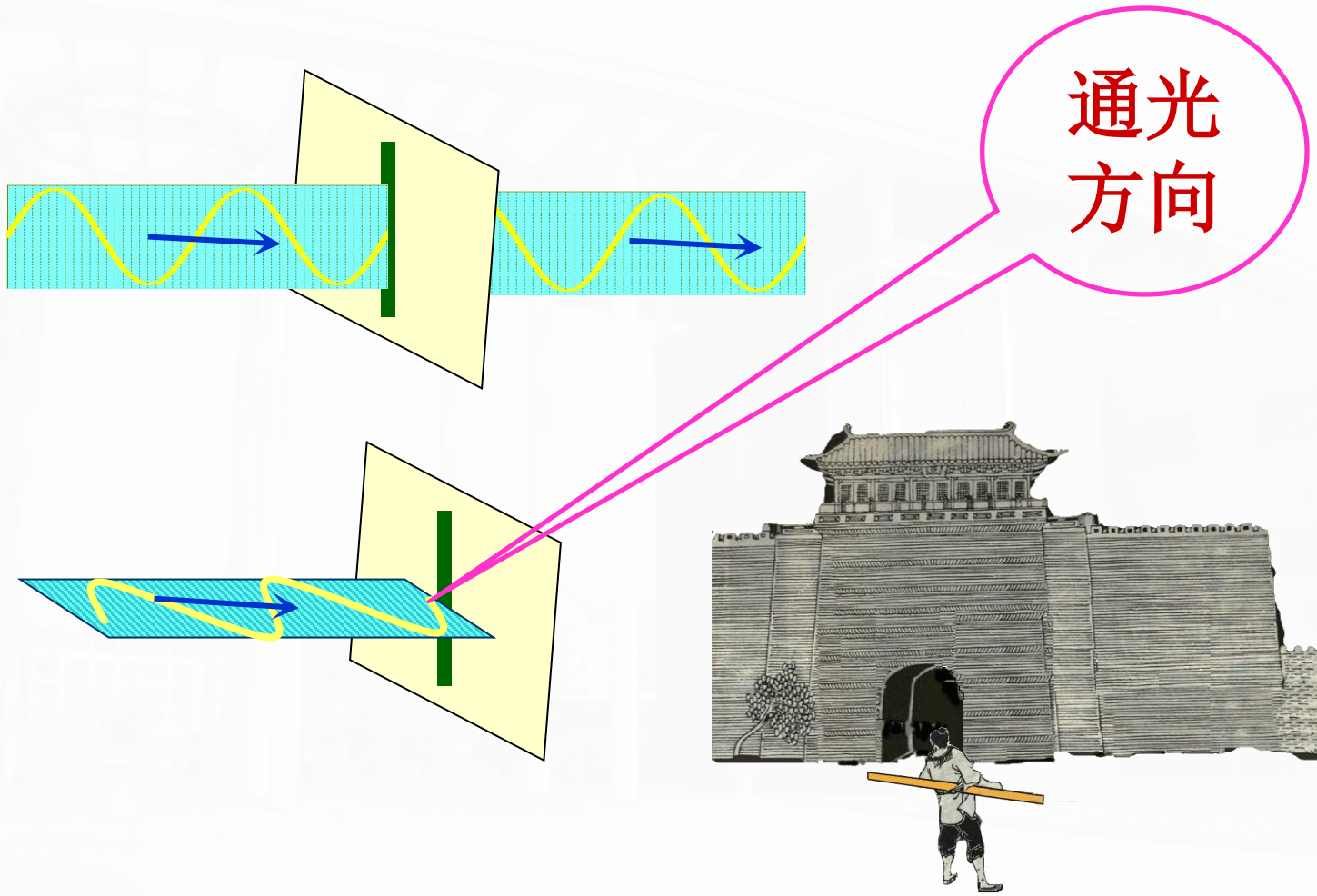
横波



纵波



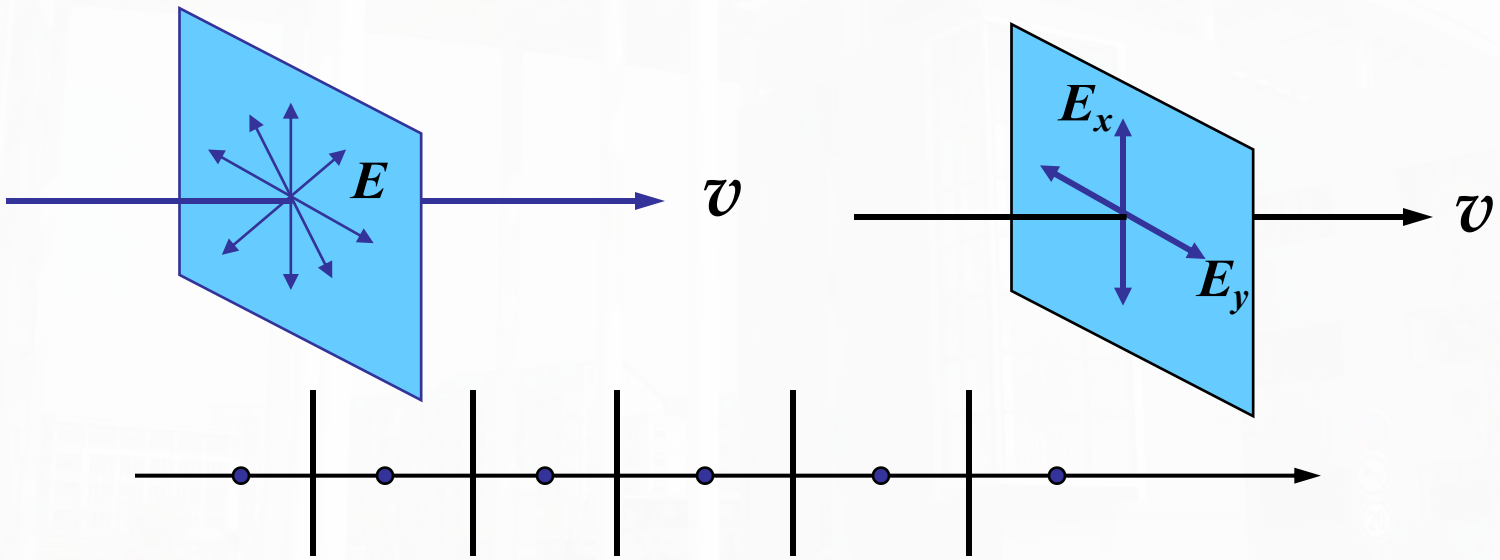
横波有偏振(polarization)现象　纵波无偏振问题



腰横扁担进不了城门

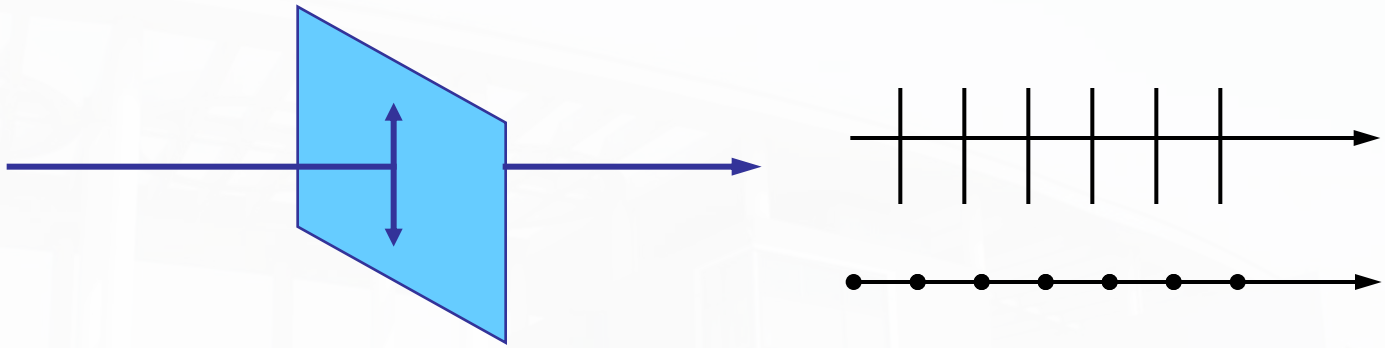
一、自然光 偏振光

因原子发光的“随机性”，光矢量各向随机均匀分布，振幅相等，称为自然光(natural light)。



各向光振动无固定相位关系。

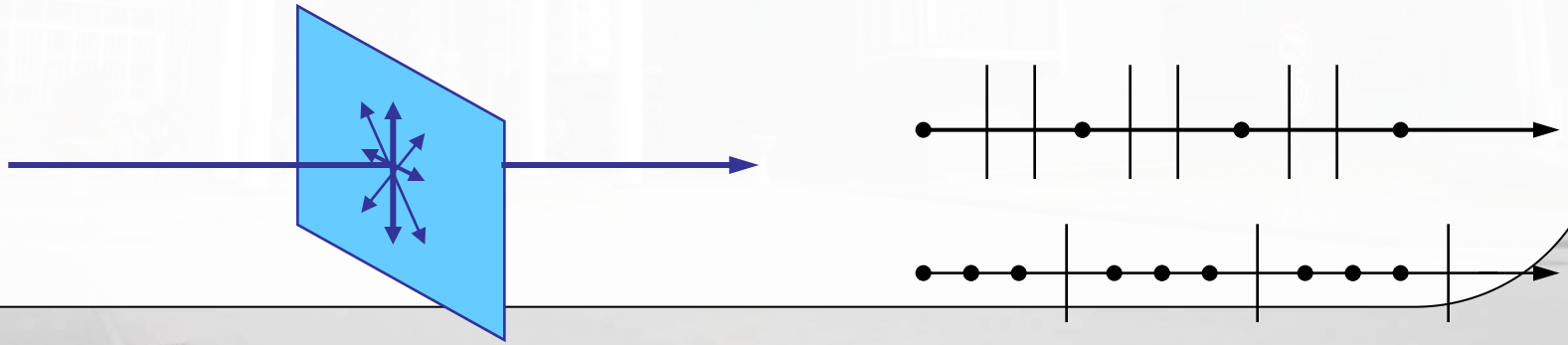
线偏振光(linearly polarized light)：光矢量只沿一个固定方向振动



振动面：光振动方向与传播方向构成的平面

部分偏振光(partial polarized light)：

一方向光振动较另其他方向占优势。线偏振光与自然光的混合。

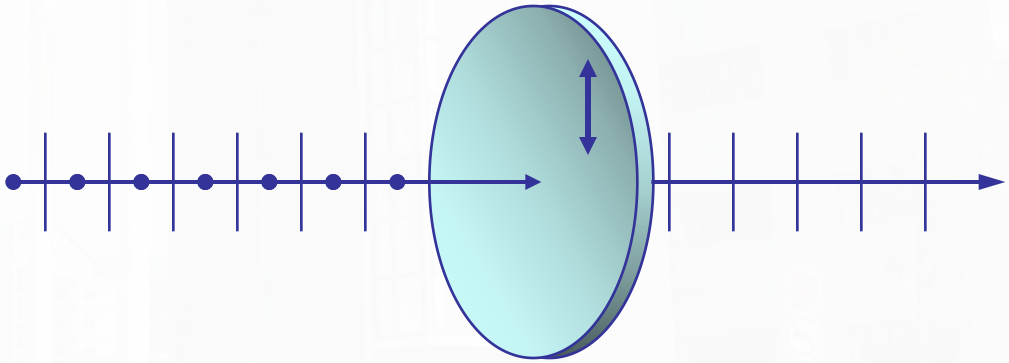


二、起偏与检偏

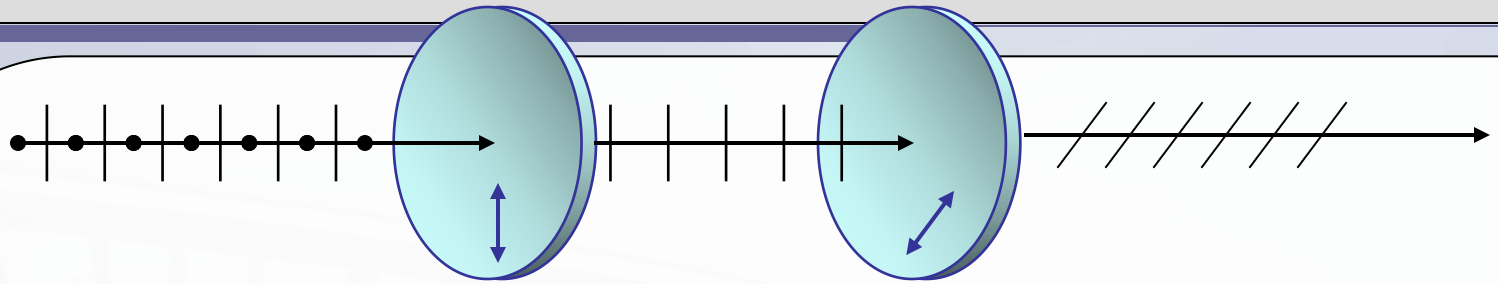
偏振片(polaroid)： 能吸收某一方向的光振动，而只让与之垂直方向上的光振动通过的一种透明薄片。

偏振化方向(polarizing direction)：

允许通过的光振动方向。



偏振片的用途： “起偏”和“检偏”

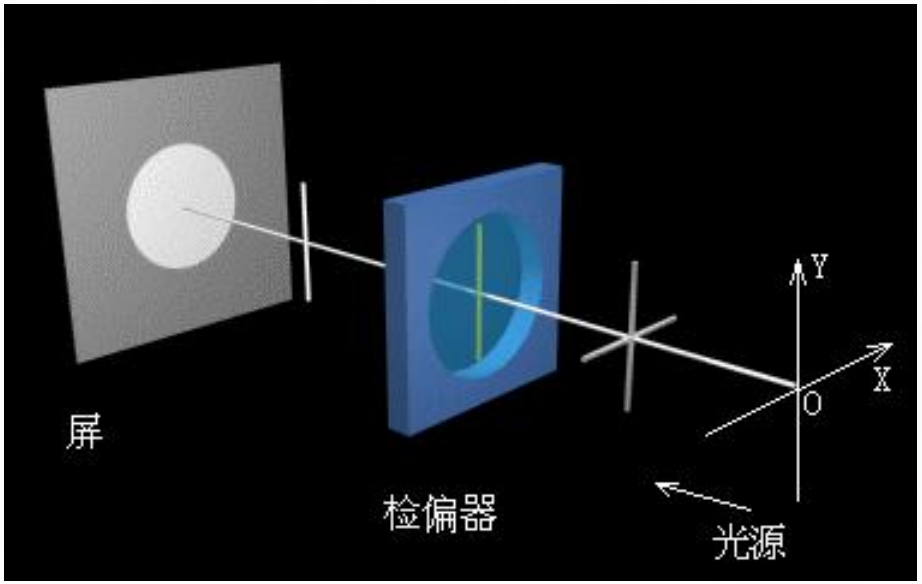


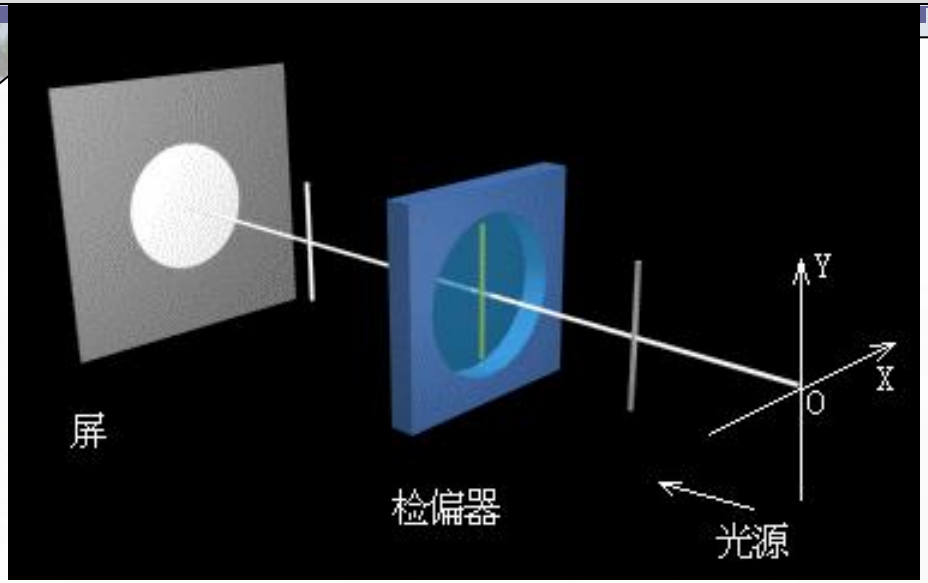
自然光 起偏 偏振光 检偏 偏振光

（1） 自然光强 I_0
→线偏振光强 ：

$$I = \frac{I_0}{2}$$

光强不变无消光





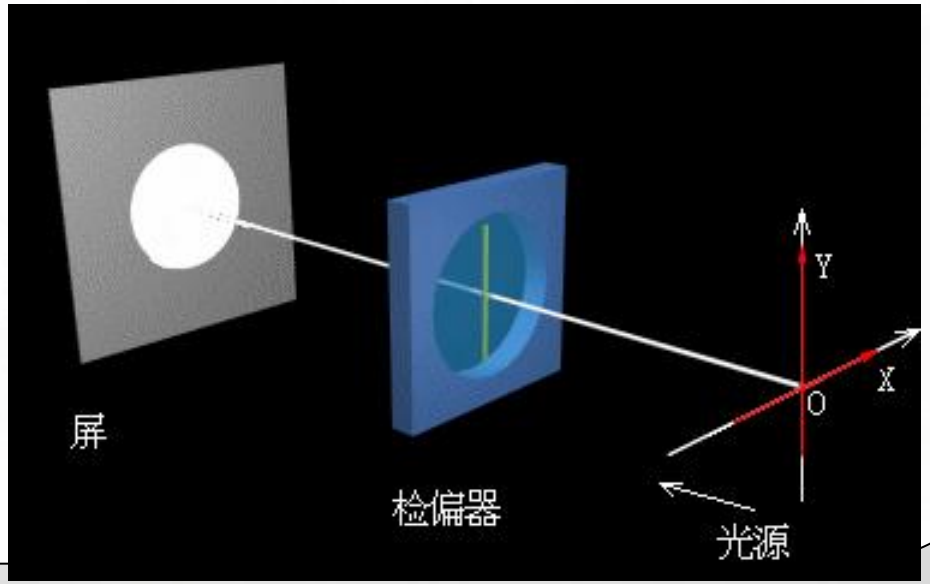
(2) 线偏振光强 I_0
→线偏振光强

$$I = I_0 \cos^2 \alpha$$

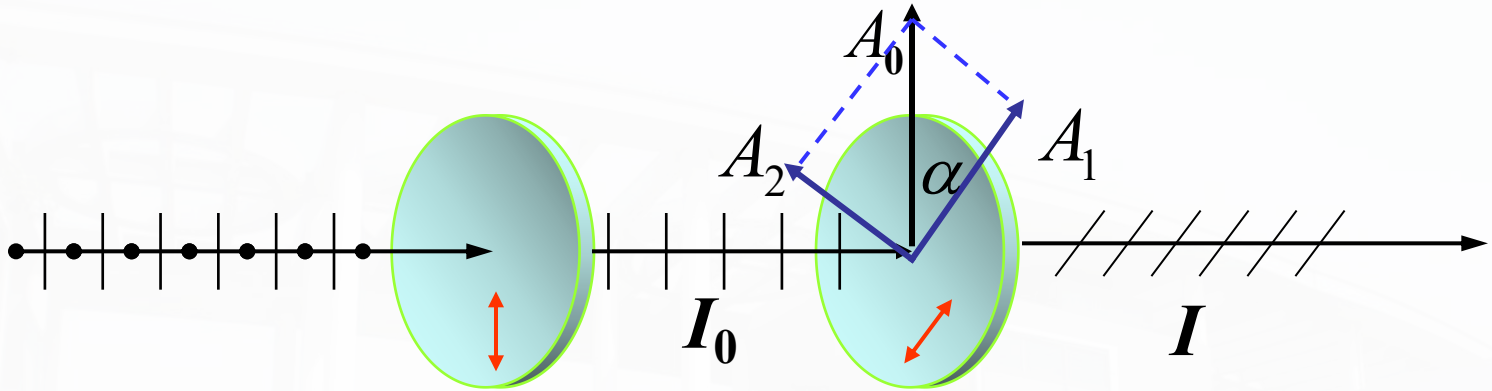
有两次消光现象

(3) 部分偏振光(混合)
→线偏振光

有两次极大极小

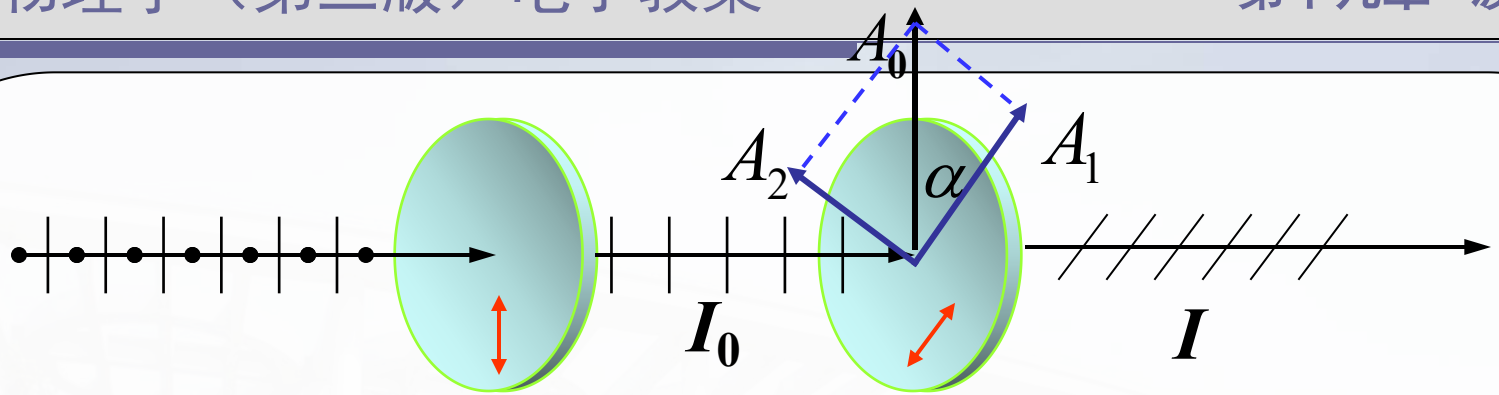


三、马吕斯定律(Malus law)



光强为I_0的线偏振光，透过偏振片后，透射强度为：

$$I = I_0 \cos^2 \alpha$$



证明：

$$A_1 = A_0 \cos\alpha$$

$$\because \quad I_0 \propto A_0^2 \quad , \quad I \propto A_1^2$$

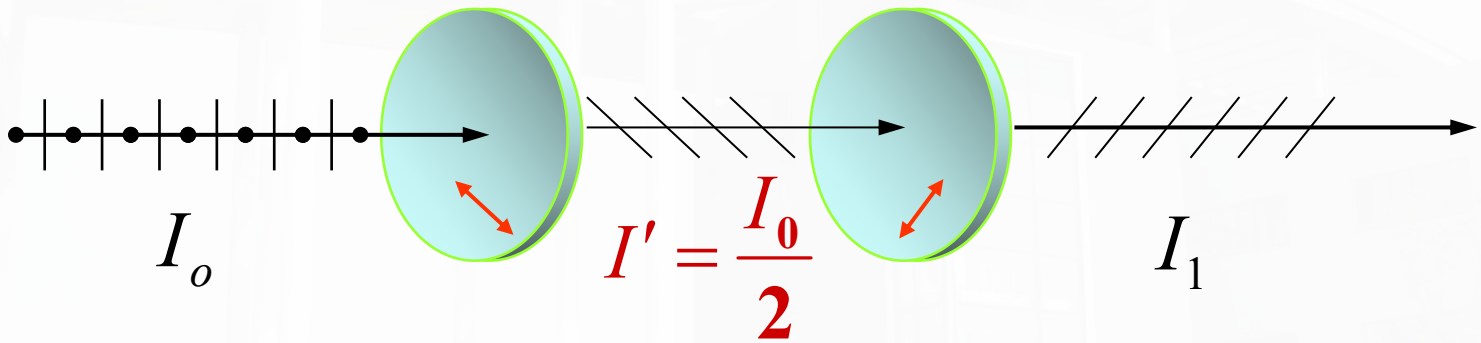
$$\therefore \quad \frac{I}{I_0} = \frac{A_1^2}{A_0^2} = \cos^2\alpha \qquad I = I_0 \cos^2\alpha$$

当： $\alpha = 0, \pi \quad \rightarrow I = I_0$

当： $\alpha = \pi/2, 3\pi/2 \quad \rightarrow I = 0$

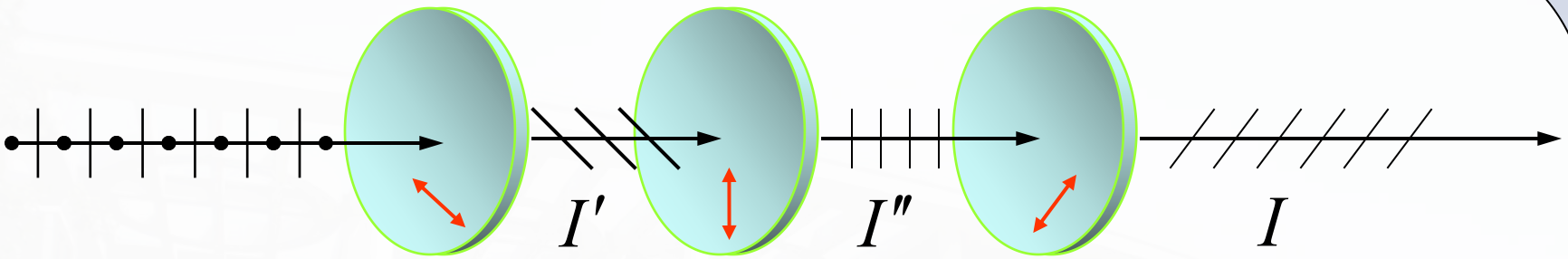
例题1 已知自然光通过两个偏振化方向相交60° 的偏振片，透射光强为I1，今在 这两偏振片之间再插入另一偏振片，它的偏振化方向与前两个偏振片的偏振化方向均夹30° 角，则透射光强为多少？

解：



$$I_1 = I'\cos^2\alpha = I'\cos^2 60° = \frac{1}{4}I'$$

$$\rightarrow I' = 4I_1$$



$$I'' = I'\cos^2 30° = 4I_1\cos^2 30° = 3I_1$$

$$I = I''\cos^2 30° = 3I_1\left(\frac{3}{4}\right) = \frac{9}{4}I_1$$

例题2 一束光由自然光和线偏振光混合组成，当它通过一偏振片时，发现透射光的强度随偏振片的转动可以变化到5倍。求入射光中自然光和线偏振光的强度各占入射光强度的几分之几？

解： 设入射光强度：I_0 ； 自然光强度：I_{10}；

偏振光强度：I_{20}

$$I_0 = I_{10} + I_{20}$$

设通过偏振片后的光强分别为：I， I_1， I_2

$$I_1 = \frac{1}{2} I_{10} \qquad I_2 = I_{20} \cos^2 \alpha$$

$$I = I_1 + I_2 = \frac{1}{2} I_{10} + I_{20} \cos^2 \alpha$$

$$\alpha = 0\text{时} \quad \rightarrow I = I_{\max} = \frac{1}{2} I_{10} + I_{20}$$

$$\alpha = 90^\circ \text{ 时} \quad \rightarrow I = I_{\min} = \frac{1}{2} I_{10}$$

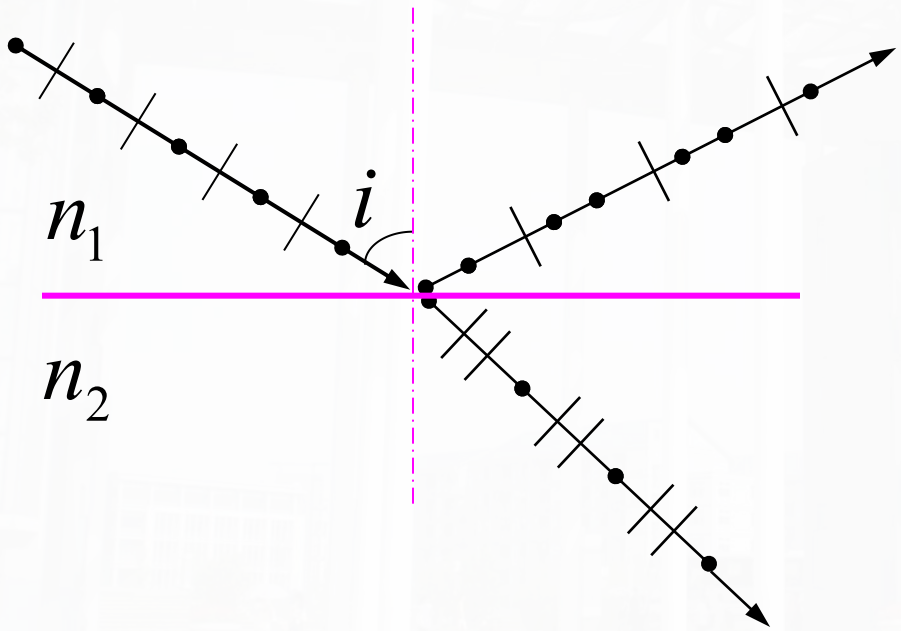
$$I_{\max} = 5 I_{\min} \qquad \rightarrow \frac{1}{2} I_{10} + I_{20} = 5 \times \frac{1}{2} I_{10}$$

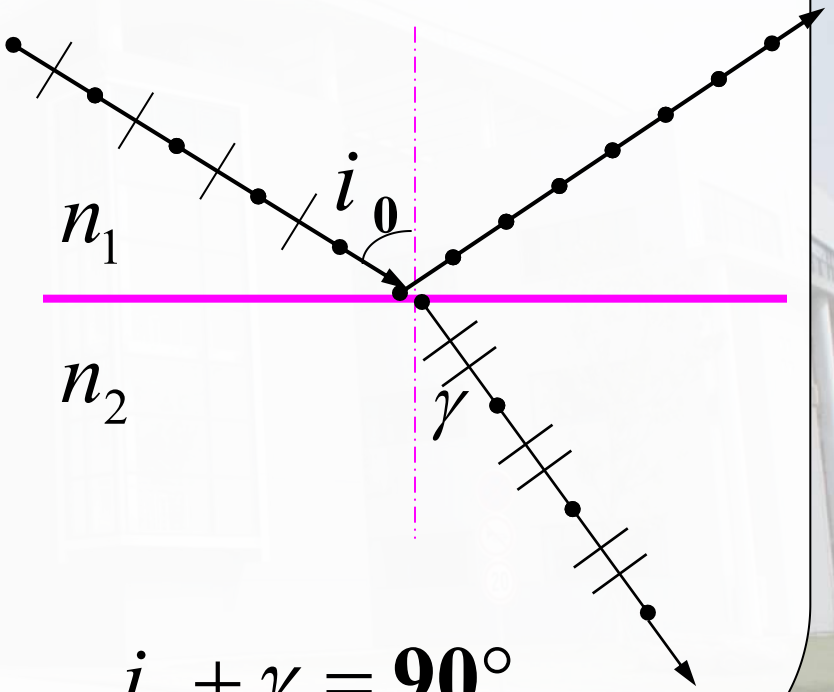
$$I_{20} = 2 I_{10}$$

$$\frac{I_{10}}{I_0} = \frac{1}{3} \qquad \frac{I_{20}}{I_0} = \frac{2}{3}$$

19－14 布鲁斯特定律

反射和折射的起偏

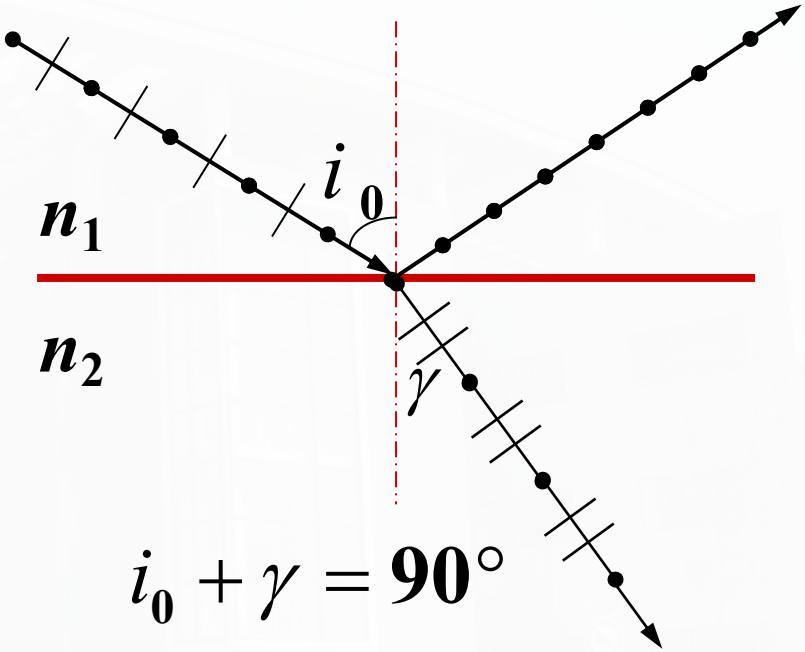




$$i_0 + \gamma = 90°$$

$$\tan i_0 = \frac{\sin i_0}{\cos i_0} = \frac{\sin i_0}{\cos(90° - \gamma)} = \frac{\sin i_0}{\sin \gamma} = \frac{n_2}{n_1}$$

布儒斯特定律: (Brewster law) 自然光以布儒斯特角入射到两不同介质表面，其反射光为线偏振光，光振动垂直于入射面。



$$i_0 + \gamma = 90°$$

布儒斯特角(polarizting angle):

$$\boxed{\tan i_0 = \frac{n_2}{n_1}}$$